

计算机图形学与虚拟现实

冼楚华

Email: chhxian@scut.edu.cn

华南理工大学计算机科学与工程学院

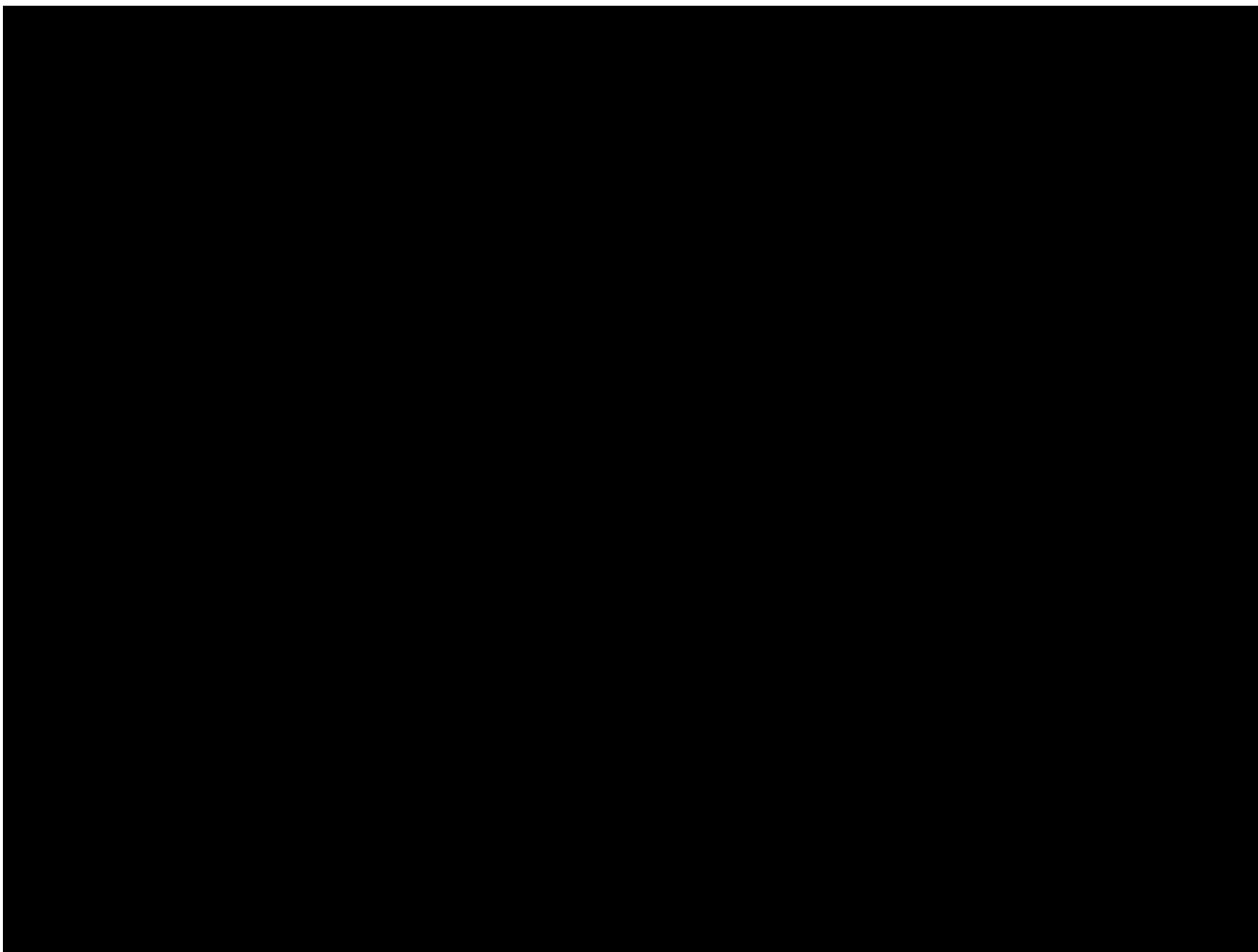
课程信息

- 授课老师姓名：冼楚华
- Email: chhxian@scut.edu.cn 
- 个人主页: <https://chuhuaxian.github.io/>
- QQ: 89071086 （比较少用，非急事请不要私聊）
- 办公室：B3-202-2
- 课程QQ群（见二维码）



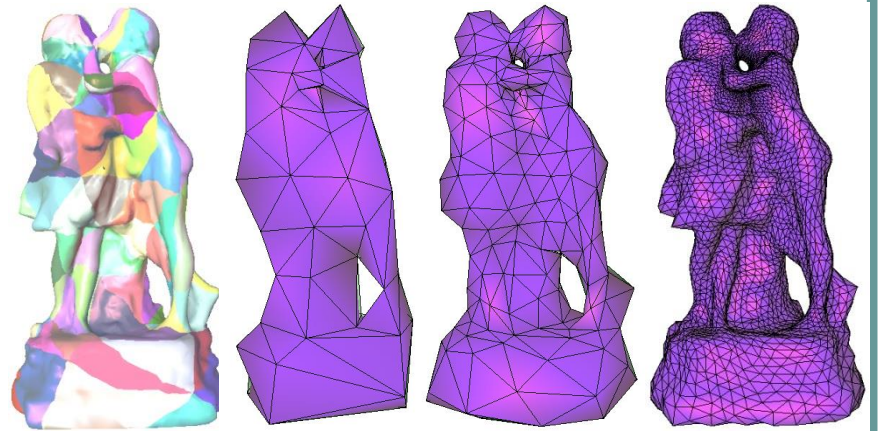
群名称：2025计算机图形学@SCUT
群 号：134660465

3D 模型

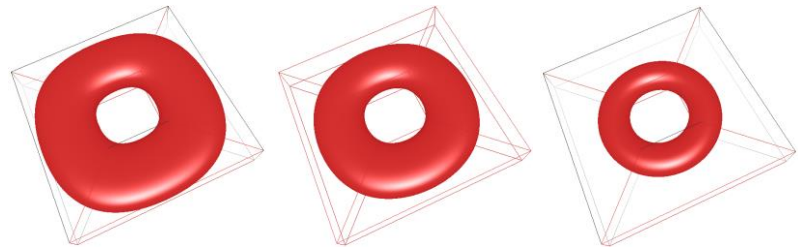


Agenda

- Meshes (网格表示)
- Subdivision curves and surfaces (细分曲面)
- L-System (L系统)
- Particle system(粒子系统)



- Parametric curves and surfaces (参数曲线曲面)



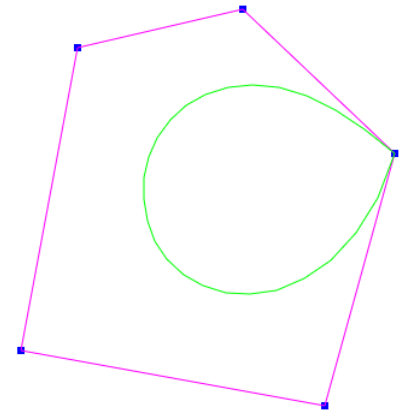
Agenda

- Parametric curves and surfaces
(参数曲线曲面)
 - Simple parametric curves and surfaces
 - Parametric curves in CAGD
 - Parametric surfaces in CAGD



Agenda

- Parametric representation of curves and surfaces
 - What is a parametric representation
 - Bezier curves and surfaces
 - B-spline curves and surfaces



直线段

- Line segment: $P_0(x_0, y_0, z_0) \rightarrow P_1(x_1, y_1, z_1)$
 - Parametric representation (参数表示)

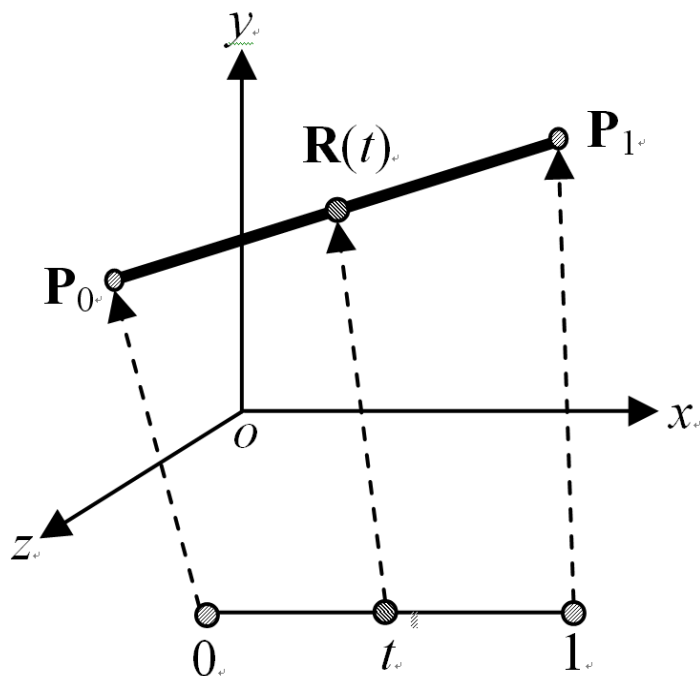
$$\mathbf{R}(t) = (1-t)\mathbf{P}_0 + t\mathbf{P}_1 \quad 0 \leq t \leq 1$$

- Coordinate components (分量表示)

$$\begin{cases} x(t) = (1-t)x_0 + tx_1 \\ y(t) = (1-t)y_0 + ty_1 \\ z(t) = (1-t)z_0 + tz_1 \end{cases} \quad 0 \leq t \leq 1$$

- Parametric domain(参数域) $[0,1]$: $0 \leq t \leq 1$

几何意义(Geometric meaning)



- Mapping from **domain** to **line segment** (映射)
- Endpoints correspond. (端点对应)

$$\mathbf{R}(0) = \mathbf{P}_0 \quad \mathbf{R}(1) = \mathbf{P}_1$$

一般参数曲线

- 一般形式

$$\mathbf{R}(t) = (x(t), y(t), z(t))$$

- Mapping: $t \leftrightarrow \mathbf{R}(t)$ (参数域 $\leftrightarrow$ 曲线上点)
- 参数域有限区间 (a finite interval) $\rightarrow$ parametric curve segment (参数曲线段)
- $x(t), y(t), z(t)$ 是分段有理多项式 (piecewise rational **polynomials**)

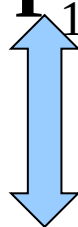
Bilinear quadrilateral patches (双线性四边形曲面片)

- Parametric expression(表达式)

$$\mathbf{R}(u, v) = (1 - v) \left[(1 - u) \mathbf{P}_0 + u \mathbf{P}_1 \right] + v \left[(1 - u) \mathbf{P}_3 + u \mathbf{P}_2 \right]$$

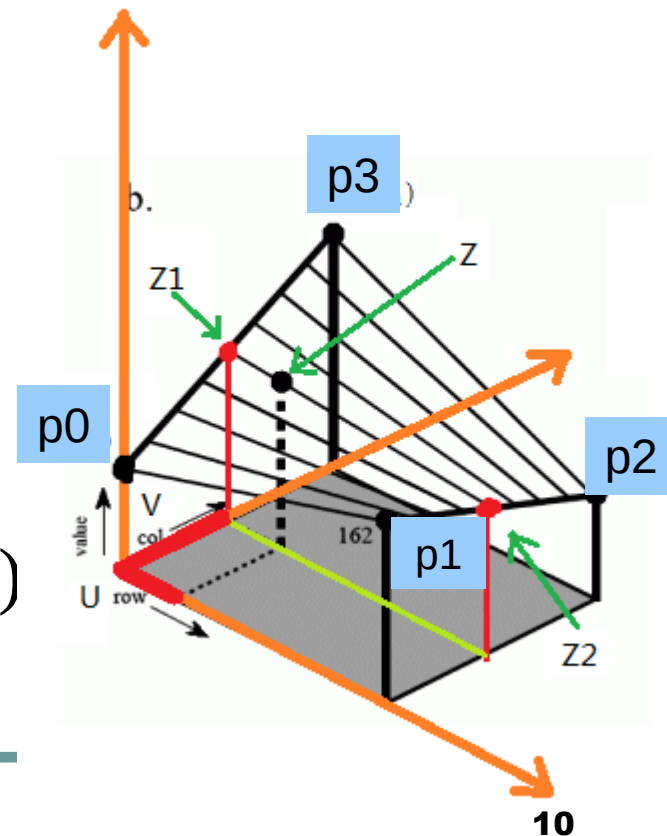
$$(u, v) \in [0, 1] \times [0, 1]$$

- Four points $\mathbf{P}_0$ 、 $\mathbf{P}_1$ 、 $\mathbf{P}_2$ and $\mathbf{P}_3$

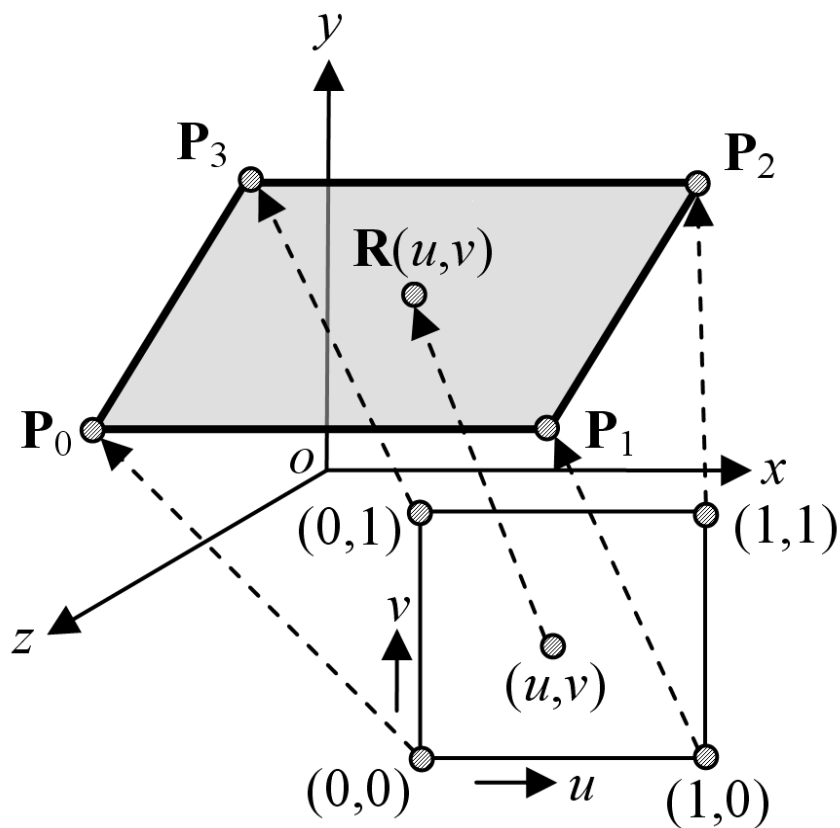


four corners of the surface:

$\mathbf{R}(0,0)$ 、 $\mathbf{R}(1,0)$ 、 $\mathbf{R}(1,1)$ and $\mathbf{R}(0,1)$



Geometric meaning(几何意义)



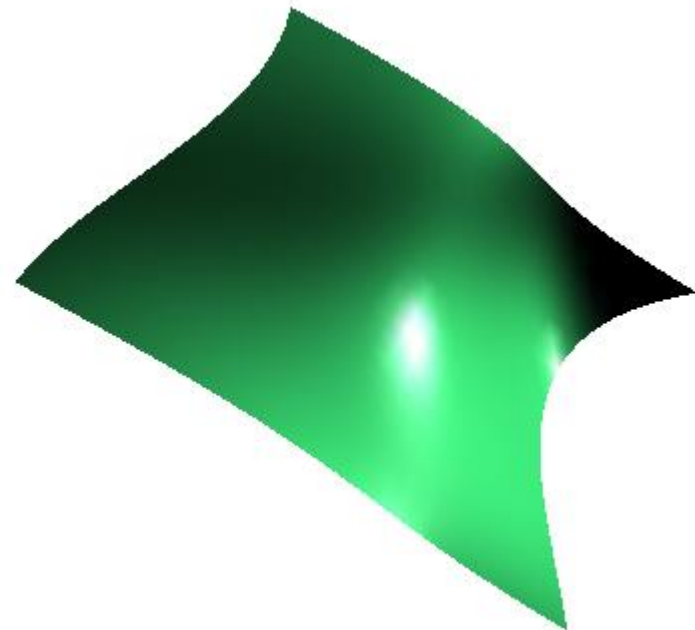
Bilinear quadrilateral patch(双线性四边形曲面片)

What are parametric surfaces: general surfaces (参数曲面一般形式)

- General form (参数曲面的一般形式)

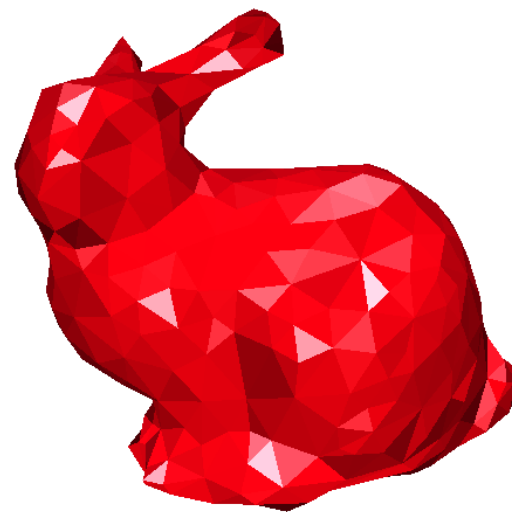
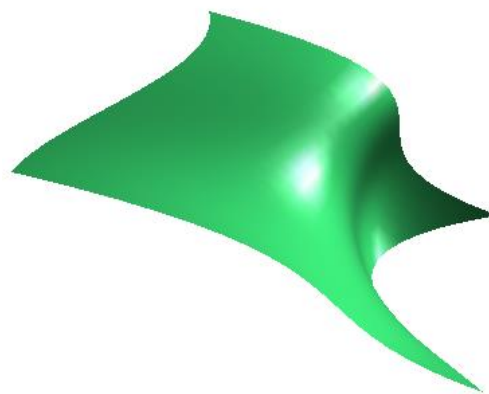
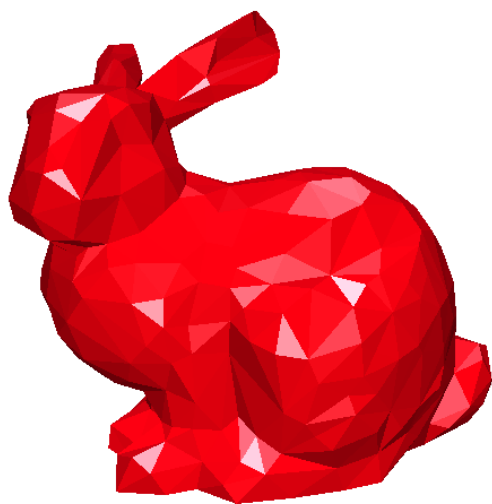
$$\mathbf{R}(u, v) = (x(u, v), y(u, v), z(u, v))$$

- One-to-one mapping: $(u, v) \leftrightarrow \mathbf{R}(u, v)$ (一一映射)
- 有限的参数域(A finite parametric domain, e.g. a rectangle)
- 多项式或有理多项式[(rational) polynomials]



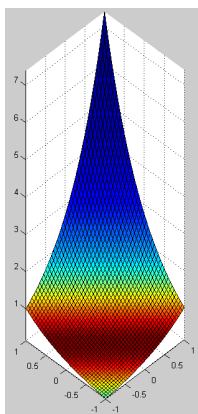
参数表示的优点

- 参数表示是显式的
 - 易于计算曲面上点的几何属性
 - 易于网格化

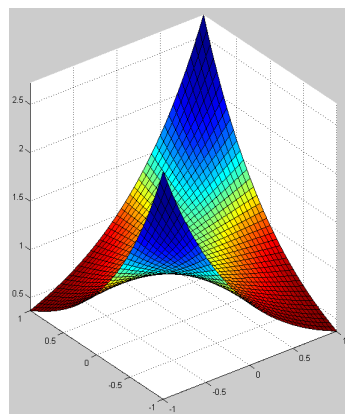


参数表示的优点

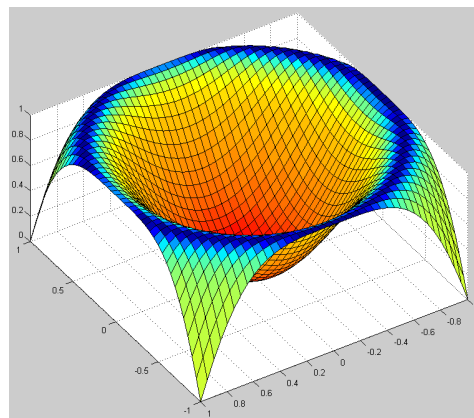
- 易于计算：法向、曲率、测地线、曲率线等



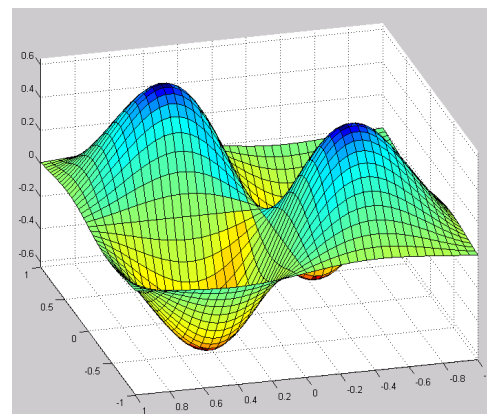
(a)



(b)



(c)

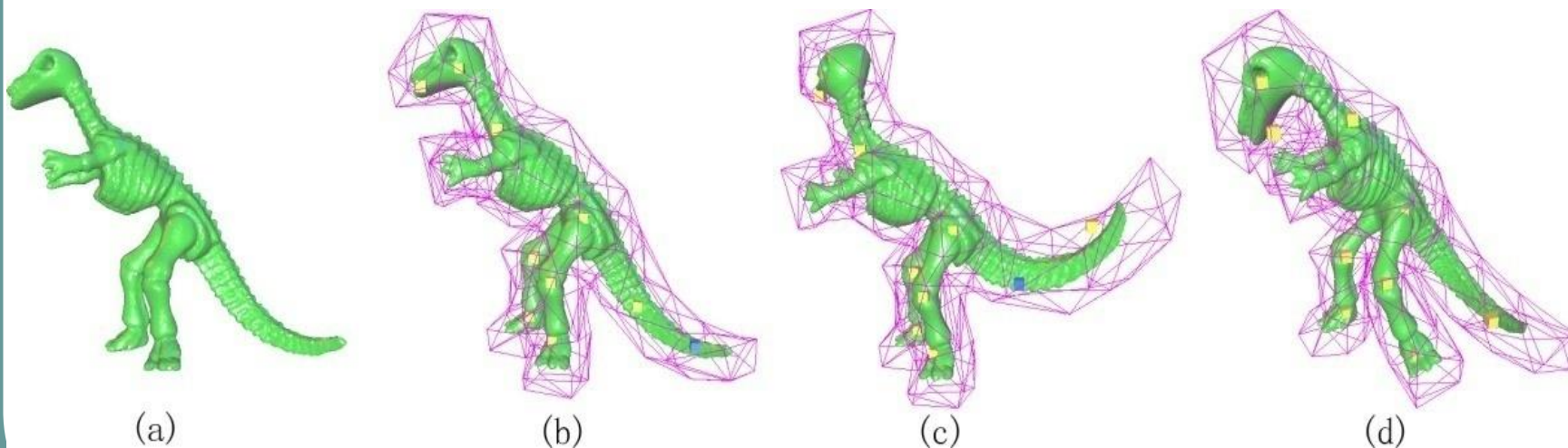


(d)

Pseudo-color visualization of mean curvature
(平均曲率伪彩绘制)

参数表示的优点

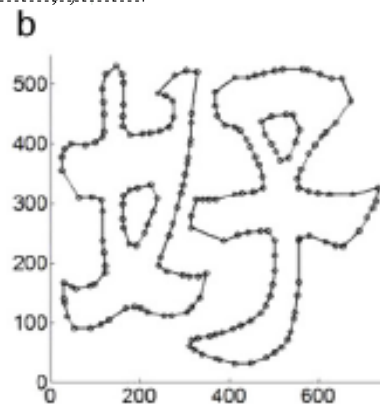
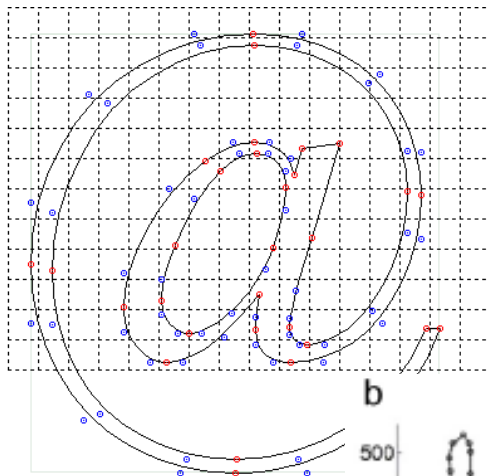
- 易于控制曲面形状
 - Bézier、B-样条、NURBS (Non-Uniform Rational B-Spline, 非均匀有理B-样条)曲线/曲面



Mesh shape editing needs to solve large scale optimization

Applications

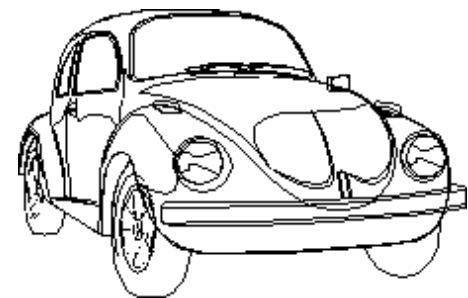
Fonts



Path planning

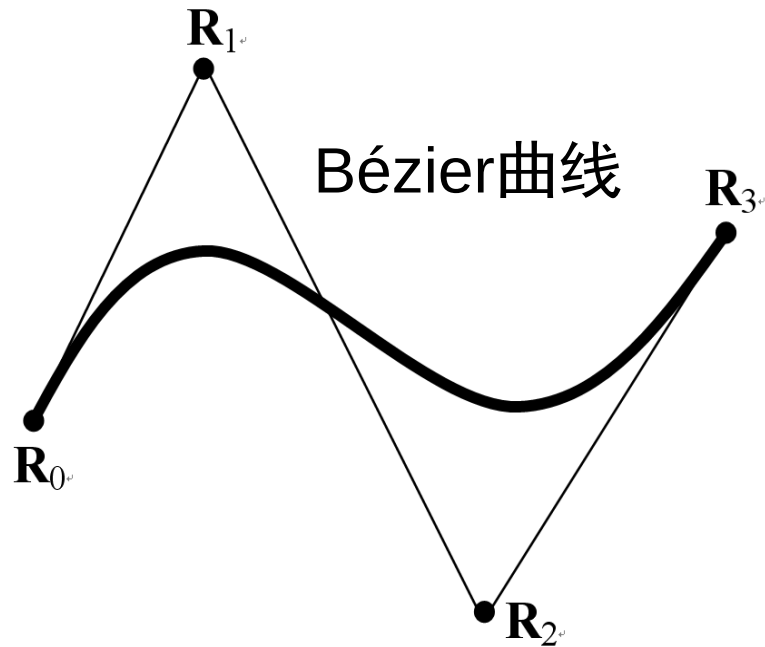


Shape design



Bézier曲线

给定四个点



Bézier曲线上得点由**控制点**混合得到。

目标：曲线要光滑；形状控制要直观

Bézier曲线



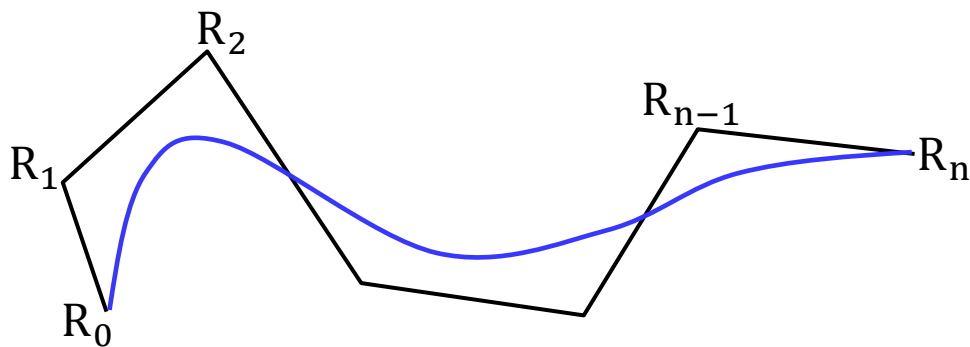
N次Bézier曲线定义

- 控制多边形: $R_0 R_1 \dots R_n (R_k = (x_k, y_k, z_k))$
- Bézier曲线公式

$$R(t) = (x(t), y(t), z(t)) = \sum_{k=0}^n R_k B_{k,n}(t), 0 \leq t \leq 1$$

$B_{k,n}(t) = \binom{n}{k} (1-t)^{n-k} t^k$ 为Bernstein基/混合函数, 满足:

$$\sum_{k=0}^n B_{k,n}(t) = (1-t+t)^n = 1$$



Bézier曲线性质

- 插值端点(Interpolating two endpoints)

- $\mathbf{R}(0) = \mathbf{R}_0 \quad \mathbf{R}(1) = \mathbf{R}_n$

- 切向量计算(Tangent vectors)

- $\mathbf{R}'(0) = n(\mathbf{R}_1 - \mathbf{R}_0)$

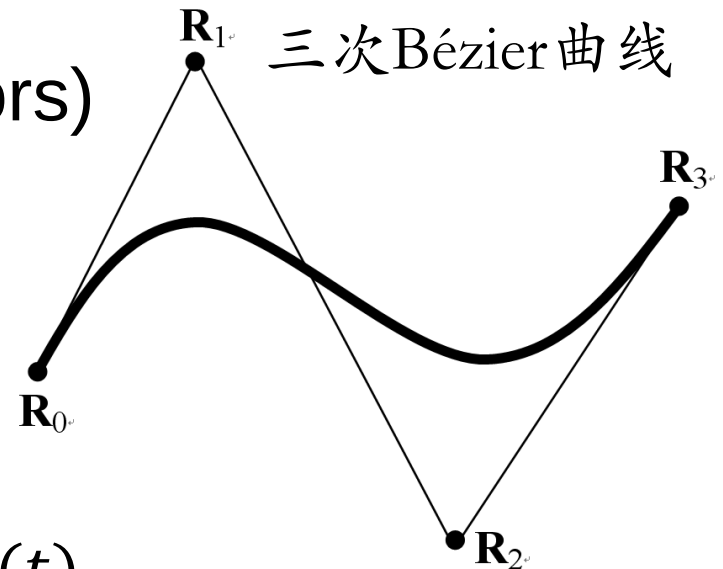
- $\mathbf{R}'(1) = n(\mathbf{R}_n - \mathbf{R}_{n-1})$

-

- 对称性(Symmetry)

- $\sum_i \mathbf{R}_{n-i} B_{i,n}(t) = \sum_i \mathbf{R}_i B_{i,n}(t)$

(曲线关于控制多边形顶点的枚举方向无关)



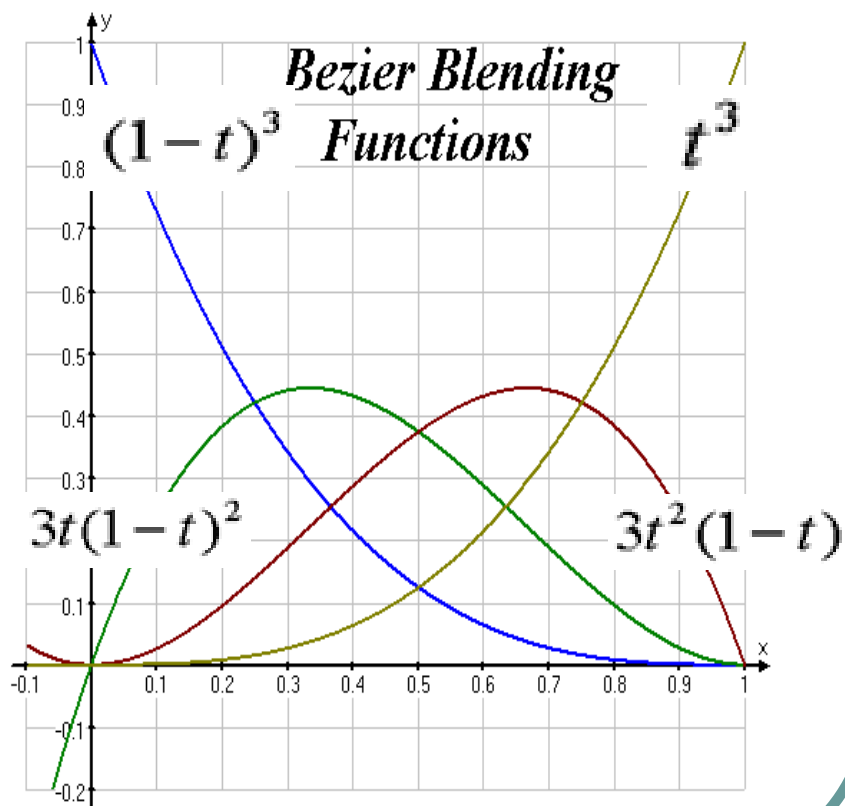
三次Bezier曲线 (Cubic Bezier Curve)

- 表示

$$R(t) = (1-t)^3 R_0 + 3t(1-t)^2 R_1 + 3t^2(1-t) R_2 + t^3 R_3$$

- 基函数取值如右图

基函数的几何形状



Bézier曲线的矩阵表示 (Bézier Matrix)

- 三次Bézier曲线为例

$$R(t) = (1-t)^3 R_0 + 3t(1-t)^2 R_1 + 3t^2(1-t) R_2 + t^3 R_3$$

$$R(t) = [t^3 \quad t^2 \quad t^1 \quad 1] \begin{pmatrix} -1 & 3 & -3 & 1 \\ 3 & -6 & 3 & 0 \\ -3 & 3 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} R_0 \\ R_1 \\ R_2 \\ R_3 \end{pmatrix}$$

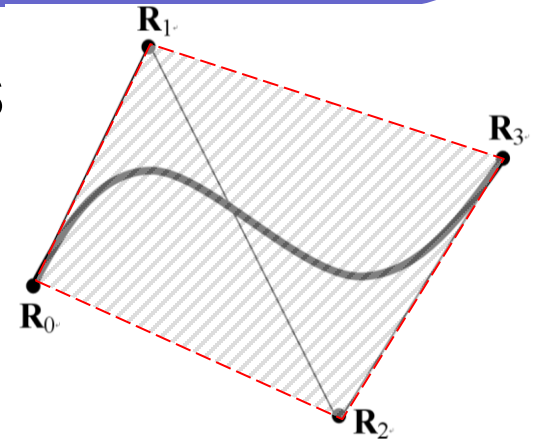
- n次Bézier曲线的矩阵表示

$$Q(t) = \mathbf{t}^T M_B \mathbf{R}$$

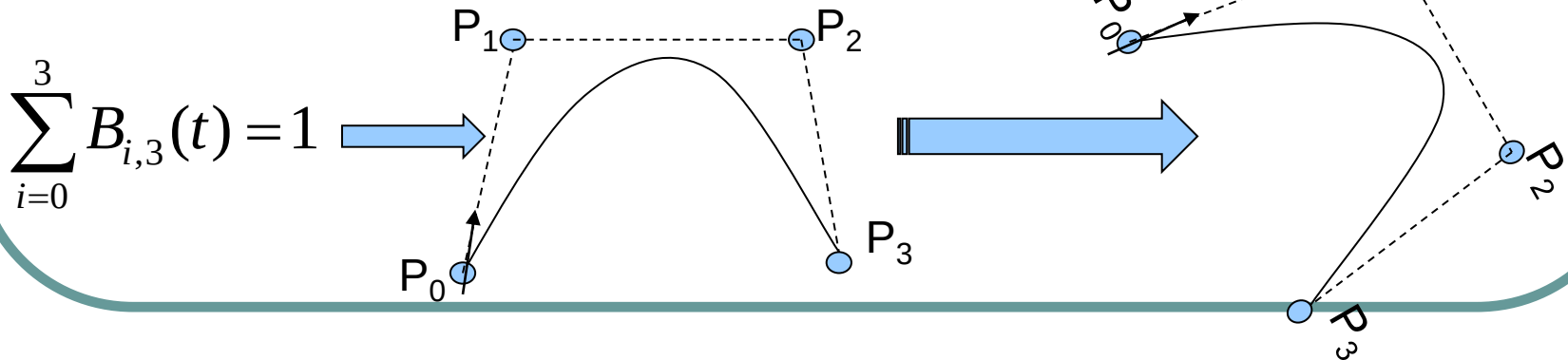
$$\mathbf{t} = (t^n \quad \dots \quad t \quad 1); M_B: \text{Bézier matrix}; \mathbf{R} = (R_0, R_1, \dots, R_n)^T$$

Bézier曲线性质(设计性质)

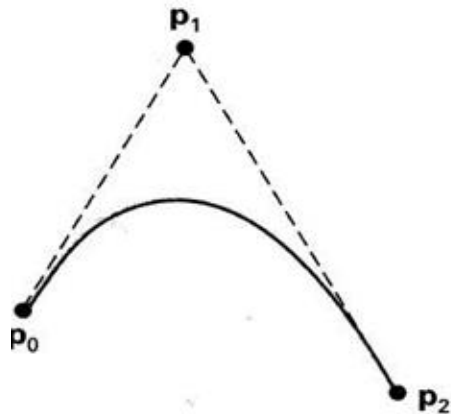
- 凸性(Convexity): Bézier curve is enveloped by the convex hull of its control polygon
- 几何不变性或仿射不变性
(Geometric invariant or affine invariant)



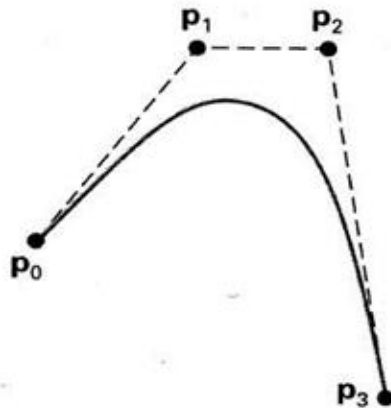
Bézier曲线的凸包性



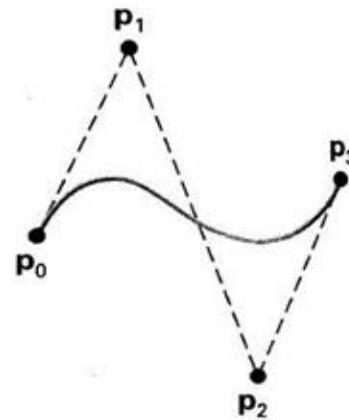
Some examples (例子)



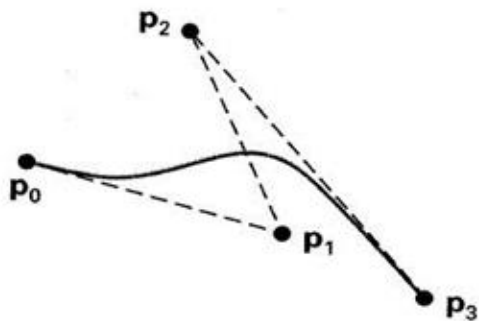
(a)



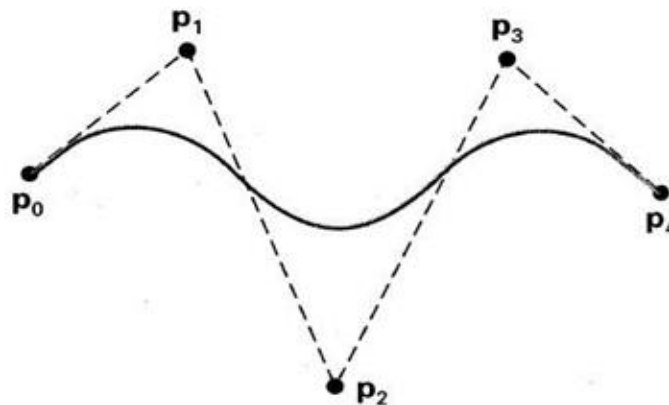
(b)



(c)



(d)



(e)

Bézier曲线细分性质

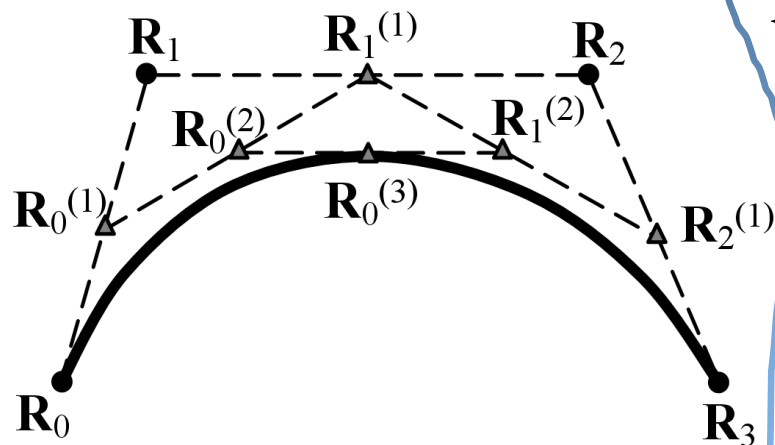


Illustration of Cubic
Bézier curve subdivision

SubdivideBezierCurve(t_0 , $\mathbf{R}$)

```
{  
    for( $i=0$ ;  $i \leq n$ ;  $i++$ )  
         $\mathbf{R}_i^{(0)} = \mathbf{R}_i$ ;  
    for( $s=1$ ;  $s \leq n$ ;  $s++$ )  
        for( $i=0$ ;  $i \leq n-s$ ;  $i++$ )  
             $\mathbf{R}_i^{(s)} = (1 - t_0) \mathbf{R}_i^{(s-1)} + t_0 \mathbf{R}_{i+1}^{(s-1)}$ ;  
    return  $\mathbf{R}(t_0)$ ;  
}
```

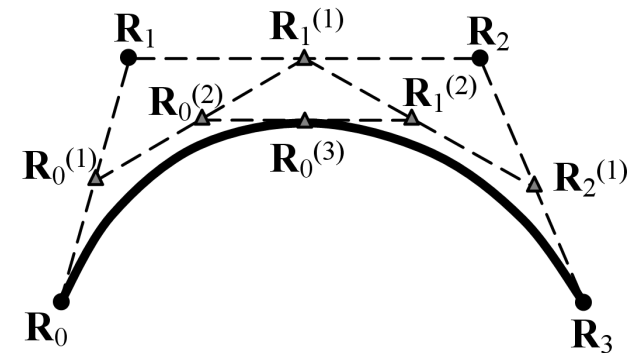
Subdivision algorithm of Bézier curve
of degree n (Bézier曲线细分算法)

Reference: http://en.wikipedia.org/wiki/B%C3%A9zier_curve

Bézier曲线的细分性质

- 一段分为两段(The curve is split into two):

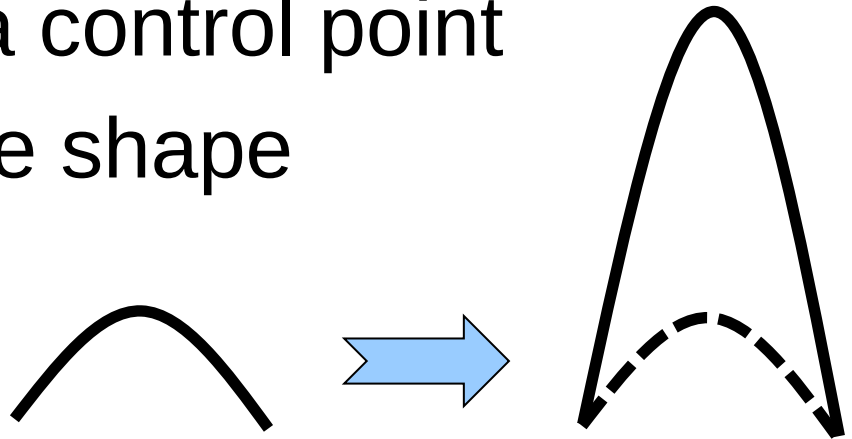
$$\left\{ \begin{array}{l} \mathbf{R}^{left}(t) = \sum_{s=0}^n \mathbf{R}_0^{(s)} B_{s,n}(t), 0 \leq t \leq 1 \\ \mathbf{R}^{right}(t) = \sum_{s=0}^n \mathbf{R}_s^{(n-s)} B_{s,n}(t), 0 \leq t \leq 1 \end{array} \right.$$



- 控制多边形越来越逼近曲线本身(The control polygon approximates the curve with the increasing of subdivision depth)
- 可用于绘制多边形(used to render the curve)

Bézier曲线缺点

- 全局性: Moving a control point changes the whole shape



- 复杂形状需拼接(stitch multiple Bezier curves)
 - 位置连续(Position continuity): C^0 (或 G^0)
 - 光滑(Smoothness): n 次导数 C^n (或几何 G^n)

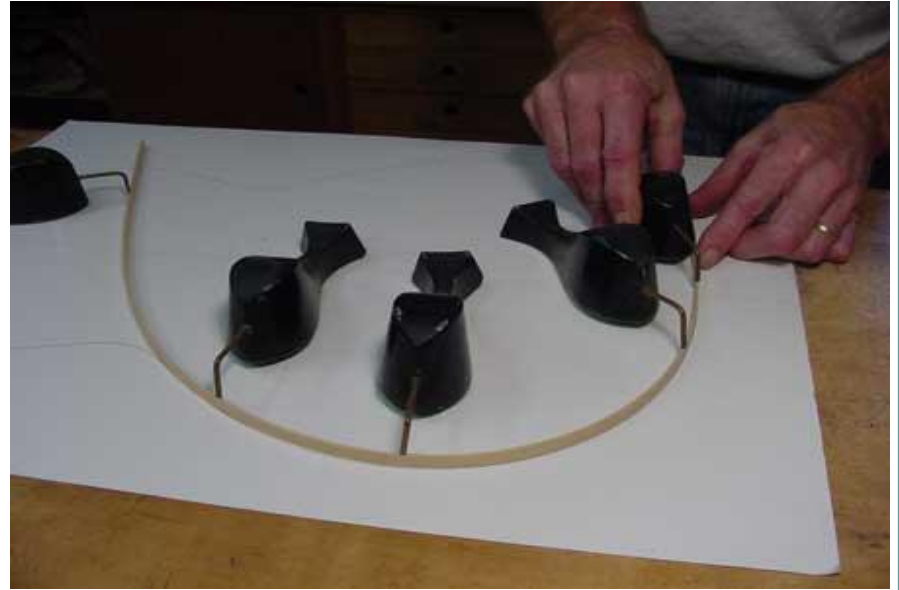
Agenda

- 参数曲线和曲面
 - 参数表示的数学原理
 - 参数曲线
 - Bézier曲线
 - B-spline (B样条曲线)
 - NURBS曲线
 - 参数曲面

历史

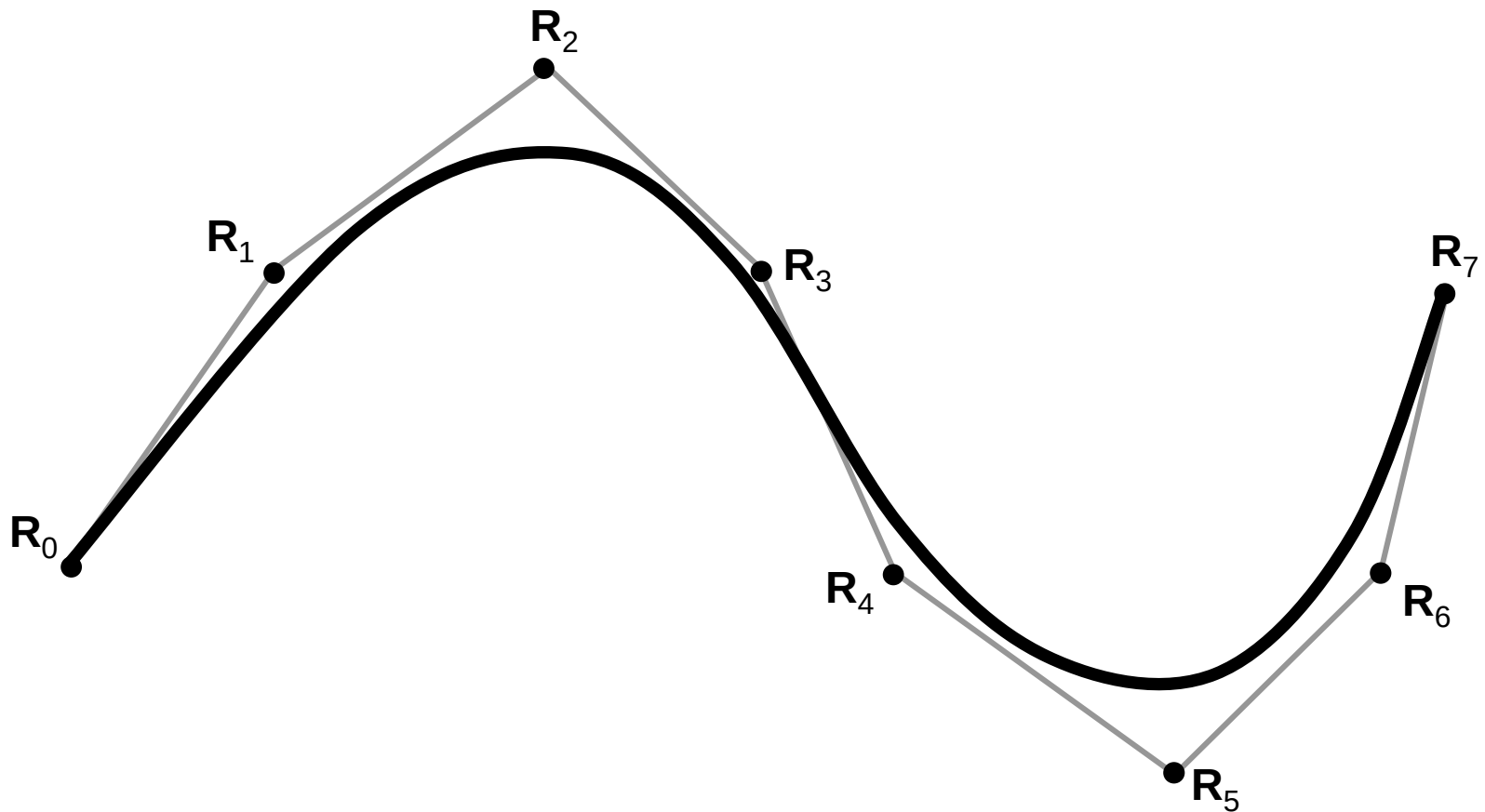


A Duck (weight)



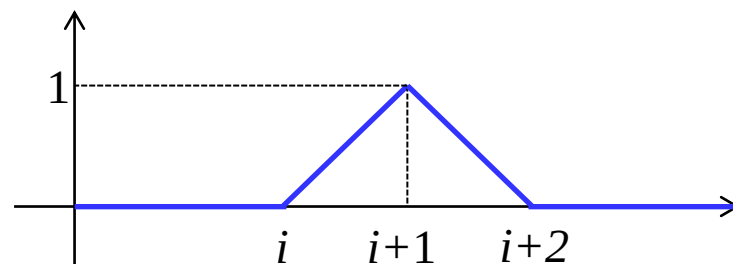
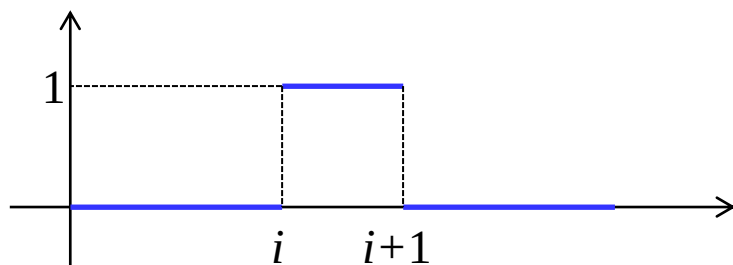
Ducks trace out curve

B-spline: example(样条曲线实例)



B-spline of degree 3(order 4) (三次(四阶)B样条曲线)

核心思想是定义分段连续多项式函数



B样条曲线定义 (B: basis)

- 节点向量(knot vector)

$$\mathbf{u} = \{u_0, u_1, \dots, u_i, \dots, u_{n+k+1}\}$$

$n + 1$ 个控制顶点, k 次 (order $k+1$) B样条

$$\mathbf{R}(u) = \sum_{i=0}^n \mathbf{R}_i N_{i,k}(u) \quad u_{k-1} \leq u \leq u_{n+1}$$

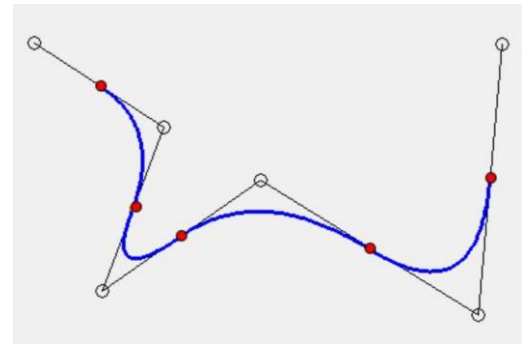
k 次或 $k + 1$ 阶 B样条

分段多项式

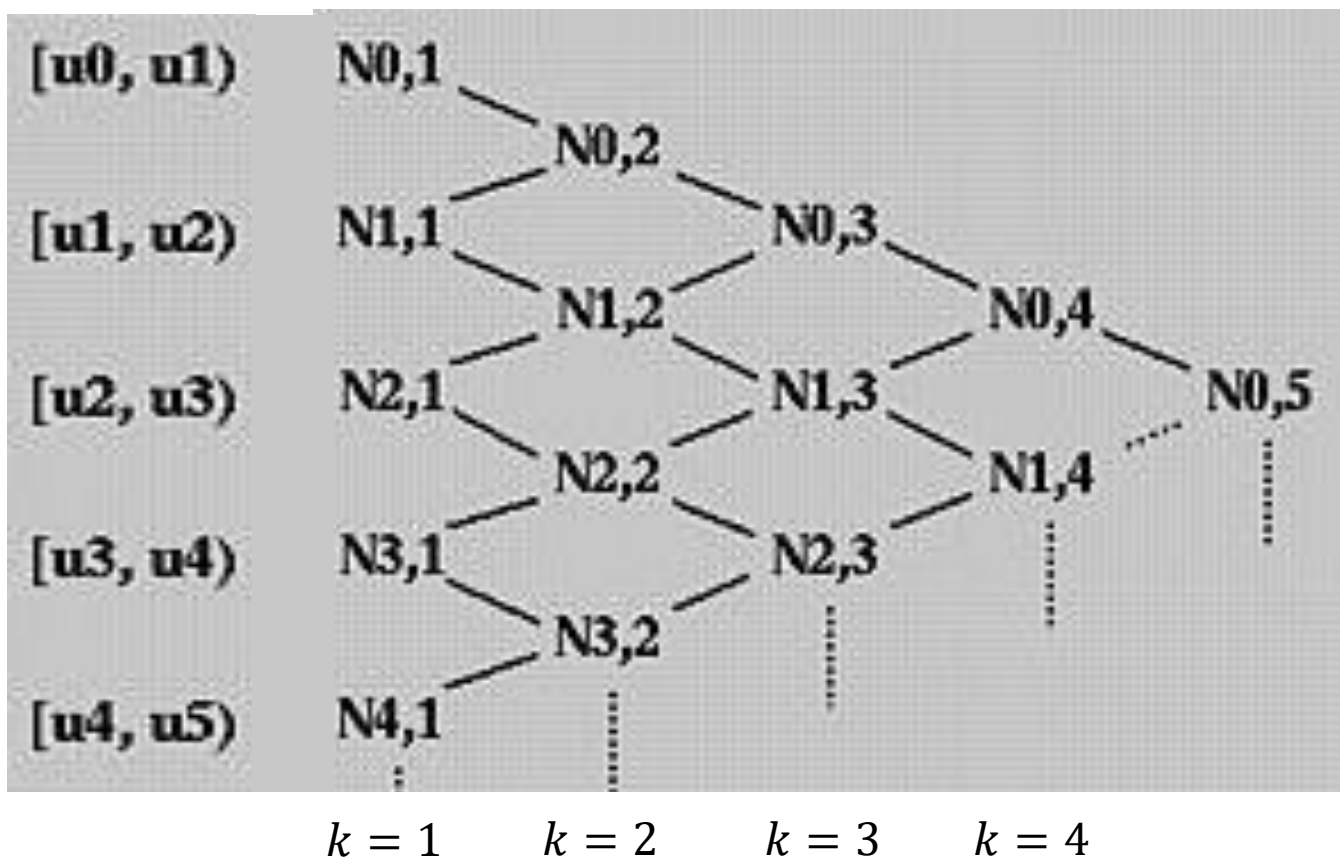
B-splines基的递归定义

- $\mathbf{R}_i$: 控制顶点, $\{\mathbf{R}_i\}_{i=0,1,\dots,n}$: 控制多边形
- $N_{i,k}(u)$ 是B样条的基(basis):

$$\left\{ \begin{array}{l} N_{i,0} = \begin{cases} 1 & \text{当 } u_i \leq u < u_{i+1} \\ 0 & \text{其它} \end{cases} \\ N_{i,k}(u) = \frac{u - u_i}{u_{i+k} - u_i} N_{i,k-1}(u) + \frac{u_{i+k+1} - u}{u_{i+k+1} - u_{i+1}} N_{i+1,k-1}(u) \\ \text{规定 } \frac{0}{0} = 0 \end{array} \right.$$



递归关系



B样条基是规范化的(normalization)

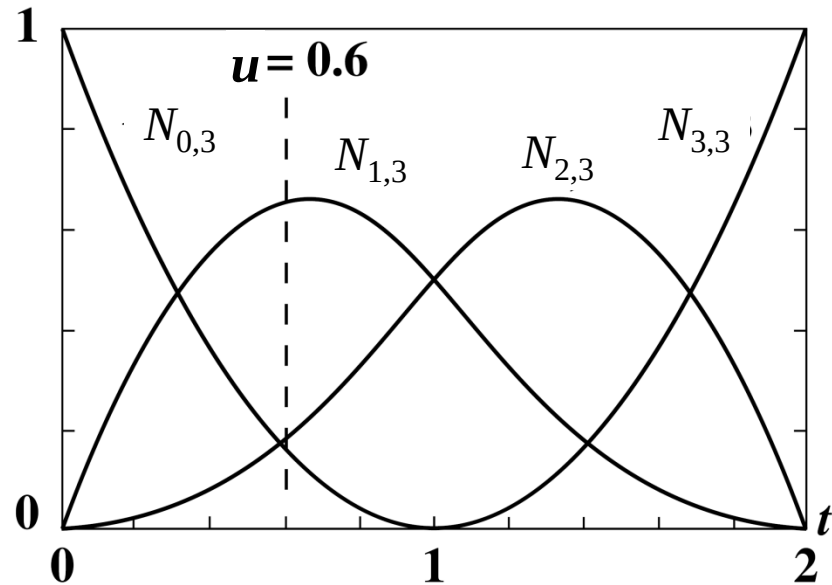
- 规范化

- 所有基的和为1

- $n=3$ (4个控制顶点)

- $k=3$ 三次(四阶)曲线

- $\mathbf{u}=[0 \ 0 \ 0 \ 1 \ 2 \ 2 \ 2 \ 2]$

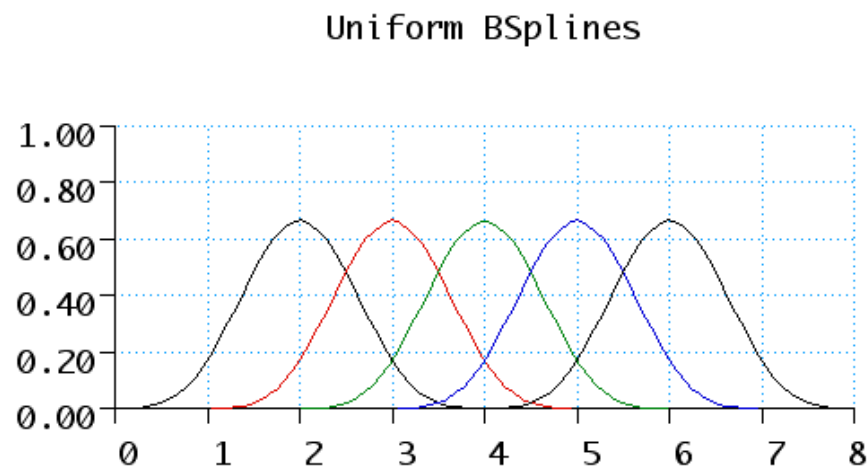


在 $u = 0.6$ 处，基函数的和为：

$$N_{0,3} + N_{1,3} + N_{2,3} + N_{3,3} = 0.16 + 0.66 + 0.18 + 0.0 = \mathbf{1.0}$$

周期均匀B样条(Periodic uniform knot)

- 周期性均匀节点向量
 - $U_i = i (0 \leq i \leq n + k + 1)$
- 例子
 - 周期均匀三次B样条
($k = 3, n = 3$) $\rightarrow$
节点数 $n + k + 2 = 8$
 - $(0, 1, 2, 3, 4, 5, 6, 7)$

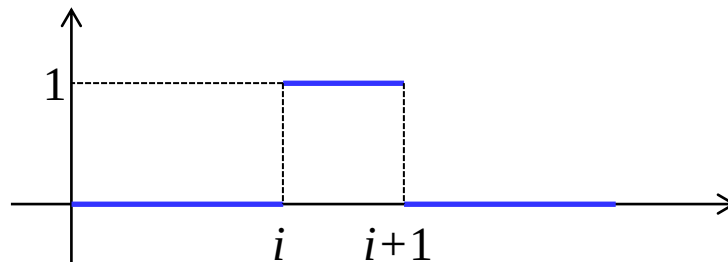


B-样条基的推导

$$\begin{cases} N_{i,0} = \begin{cases} 1 & \text{当 } u_i \leq u < u_{i+1} \\ 0 & \text{其它} \end{cases} \\ N_{i,k}(u) = \frac{u - u_i}{u_{i+k} - u_i} N_{i,k-1}(u) + \frac{u_{i+k+1} - u}{u_{i+k+1} - u_{i+1}} N_{i+1,k-1}(u) \\ \text{定义 } \frac{0}{0} = 0 \end{cases}$$

● k=0

$$N_{i,0} = \begin{cases} 1 & \text{当 } i \leq u < i+1 \\ 0 & \text{其它} \end{cases}, \quad i = 0, 1, \dots, 6$$

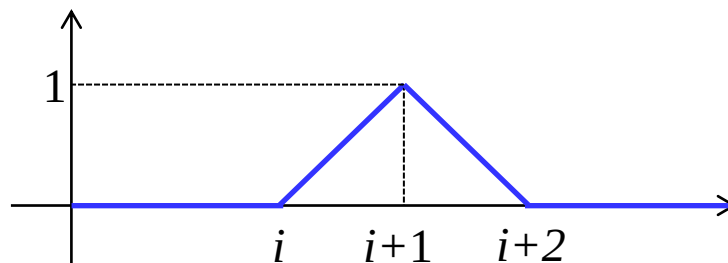


● k=1, $u_{i+k} - u_i = u_{i+k+1} - u_{i+1} = 1$

$$N_{i,1} = (u - i)N_{i,0}(u) + (i + 2 - u)N_{i+1,0}(u)$$

$$= \begin{cases} u - i, & i \leq u < i+1 \\ 2 + i - u, & i+1 \leq u < i+2 \end{cases}$$

$$i = 0, 1, 2, \dots, 5$$

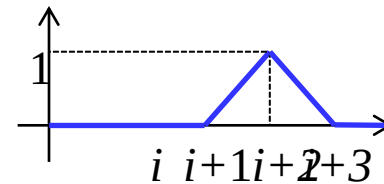
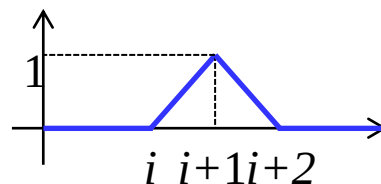


B-样条基的推导k=2

- $k=1 \quad N_{i,1} = (u-i)N_{i,0}(u) + (i+2-u)N_{i+1,0}(u)$

$$= \begin{cases} u-i, & i \leq u < i+1 \\ 2+i-u, & i+1 \leq u < i+2 \end{cases}$$

$$i = 0, 1, 2, \dots, 5$$



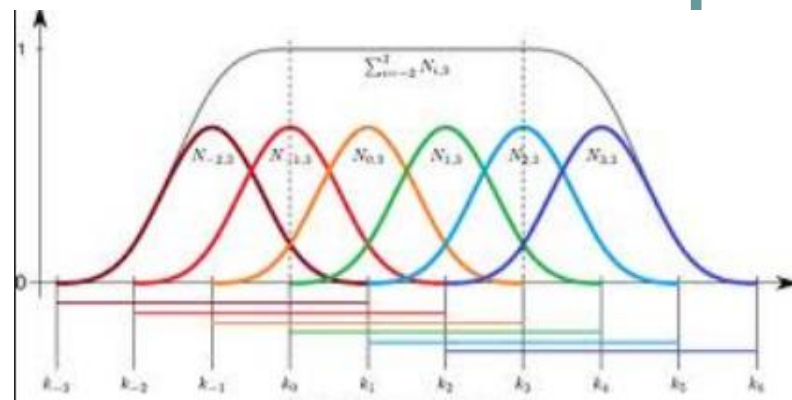
- $k=2$ 递推式:
$$N_{i,2} = \frac{u-i}{2} N_{i,1}(u) + \frac{i+3-u}{2} N_{i+1,1}(u)$$

分三段表示:

$$N_{i,2} = \frac{1}{2}(u-i)^2, \quad i \leq u < i+1$$

$$N_{i,2} = \frac{1}{2}(u-i)(i+2-u) + \frac{1}{2}(i+3-u)(u-i-1), \quad i+1 \leq u < i+2$$

$$N_{i,2} = \frac{1}{2}(i+3-u)(3+i-u), \quad i+2 \leq u < i+3$$



B-样条基的推导k=3?

- k=3?
$$N_{i,3} = \frac{u-i}{3} N_{i,2}(u) + \frac{i+4-u}{3} N_{i+1,2}(u)$$

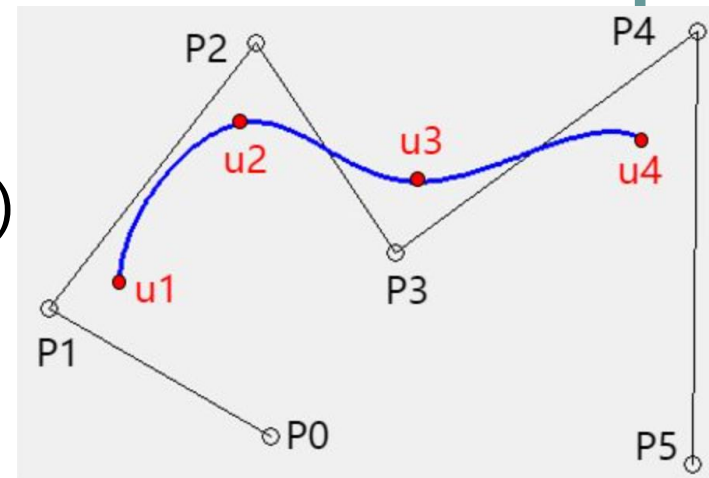
三次周期性均匀B样条：一段

- Normalize u ($0 \leq u \leq 1$, 规范化参数域)

- $N_{0,3}(u) = 1/6 (1-u)^3$
- $N_{1,3}(u) = 1/6 (3u^3 - 6u^2 + 4)$
- $N_{2,3}(u) = 1/6 (-3u^3 + 3u^2 + 3u + 1)$
- $N_{3,3}(u) = 1/6 u^3$

- 对应的参数曲线

$$R(u) = N_{0,4}(u)R_0 + N_{1,4}(u)R_1 + N_{2,4}(u)R_2 + N_{3,4}(u)R_3$$



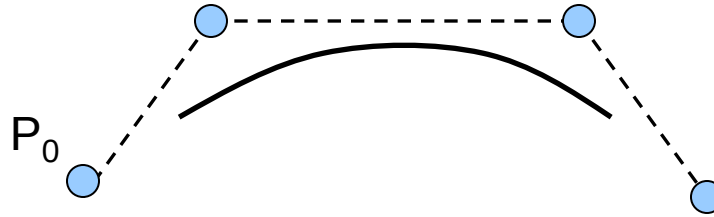
Periodic uniform cubic spline: Matrix

- In matrix form (矩阵表示)

$$R(u) = \begin{bmatrix} u^3 & u^2 & u^1 & 1 \end{bmatrix} M_n \begin{bmatrix} R_0 \\ R_1 \\ R_2 \\ R_3 \end{bmatrix}$$

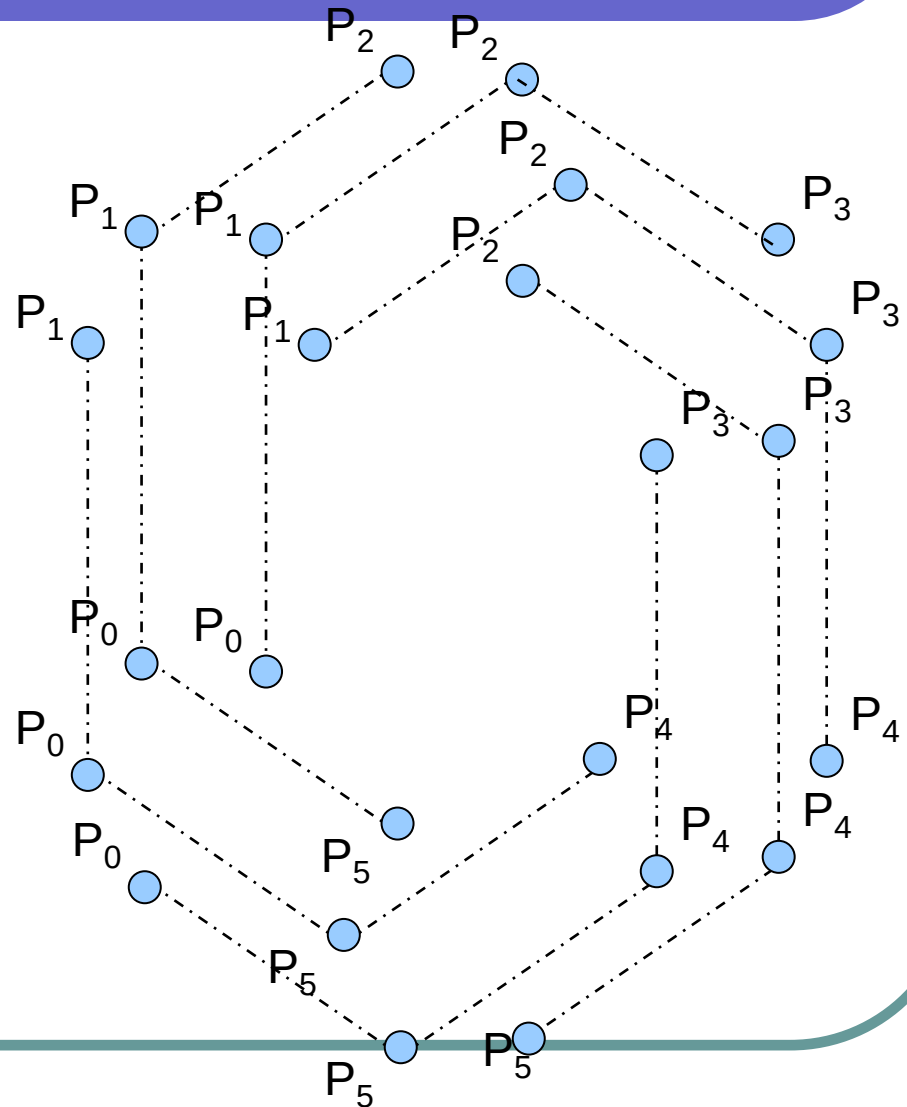
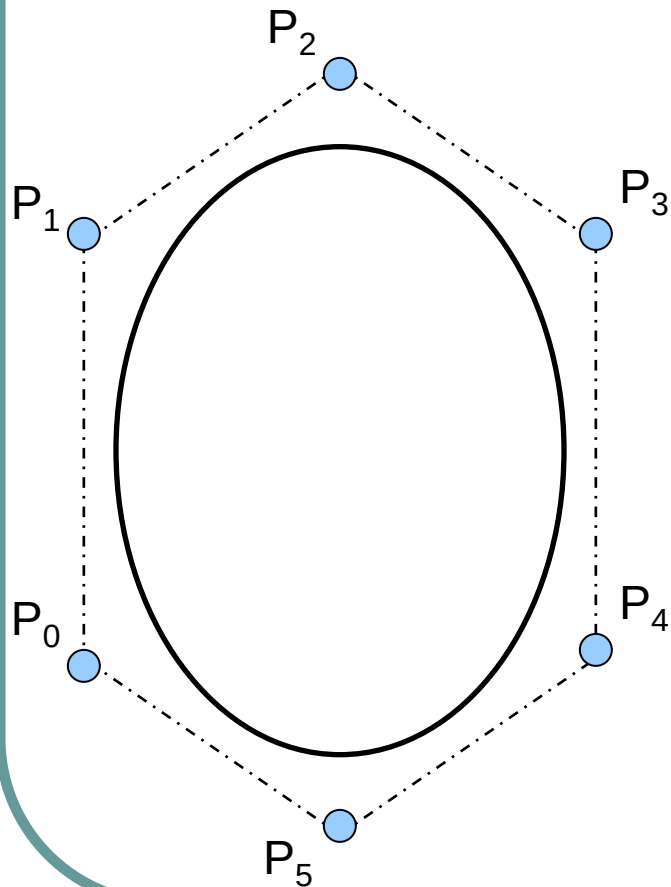
$$M_n = \frac{1}{6} \begin{bmatrix} -1 & 3 & -3 & 1 \\ 3 & -6 & 3 & 0 \\ -3 & 0 & 3 & 0 \\ 1 & 4 & 1 & 0 \end{bmatrix}$$

Periodic uniform knot



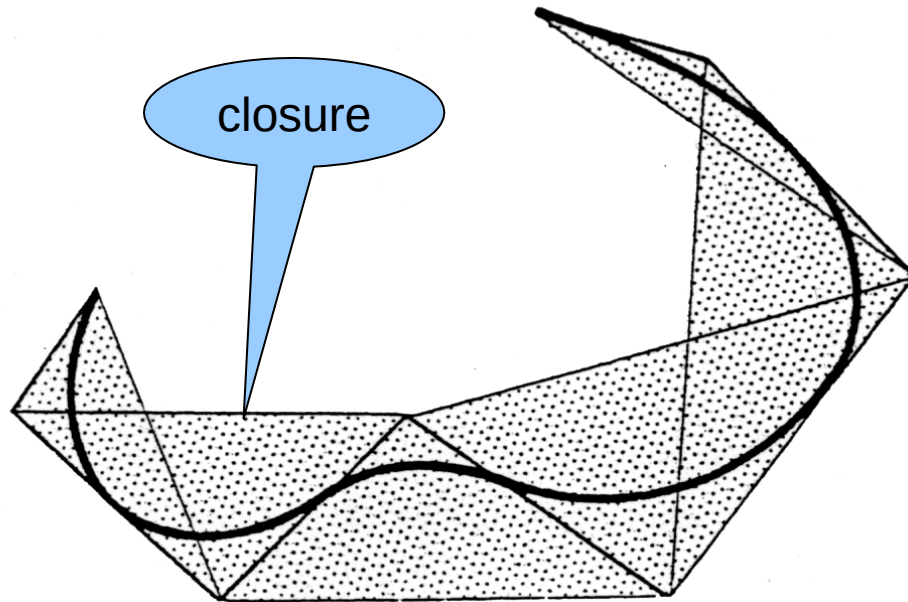
Closed periodic

Example: $k = 3, n = 7$;
 $u = \{0, 1, \mathbf{2, 3, 4, 5, 6, 7, 8}, 9, 10\}$



Properties of B-splines (性质)

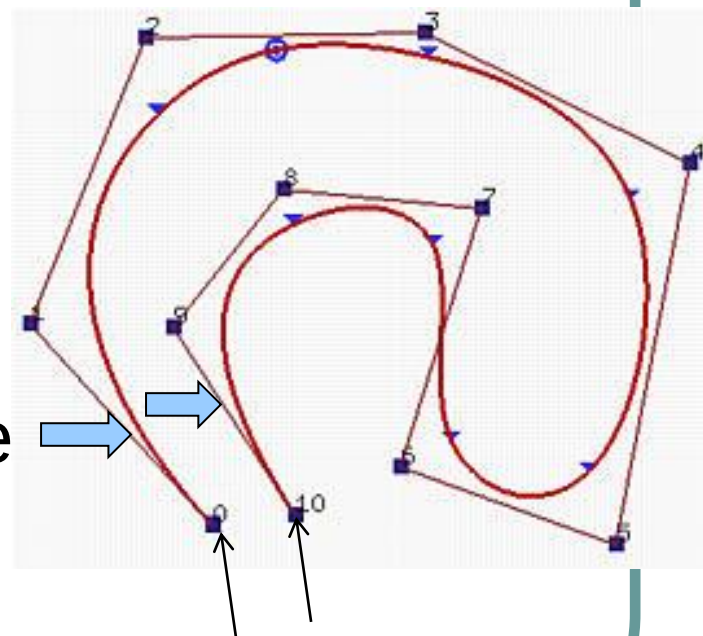
- 凸包性(Convex hull)
 - More strict than Bezier curves
- 几何不变性(Geometric invariant properties)



Quadratic B-spline (二次B样条)

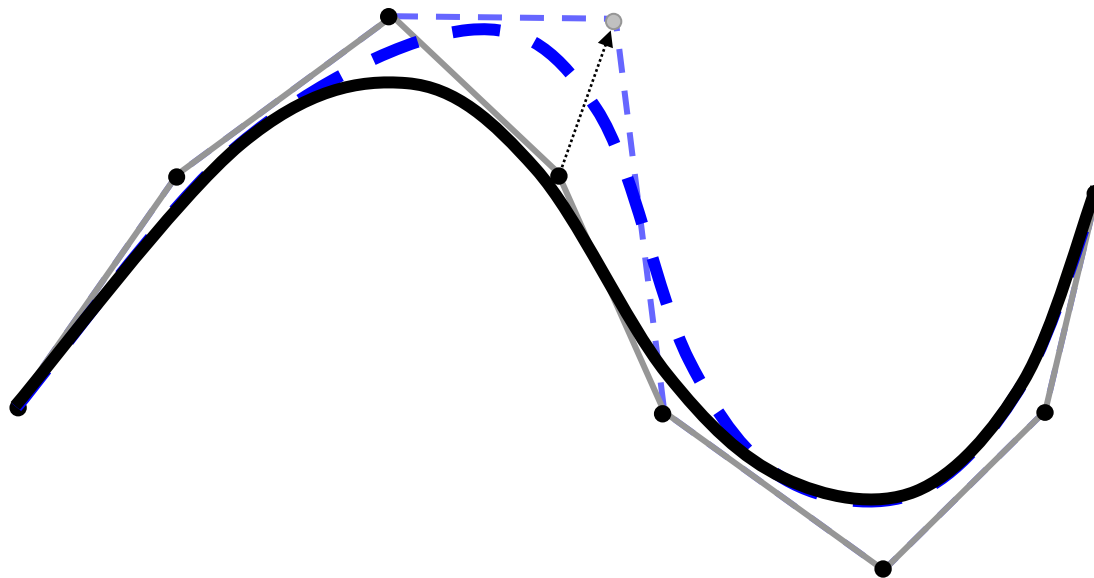
Open B-splines(非周期均匀B样条)

- 节点向量在端点处重复 $k+1$ 次
 - B样条曲线会插值端点
 - 端点处切向量与边一致
- For $n=k+1$, the B-spline degenerates to a Bézier curve



(Continued)

- **局部性(Local property)**
 - 移动一个控制顶点，只影响曲线的局部区域



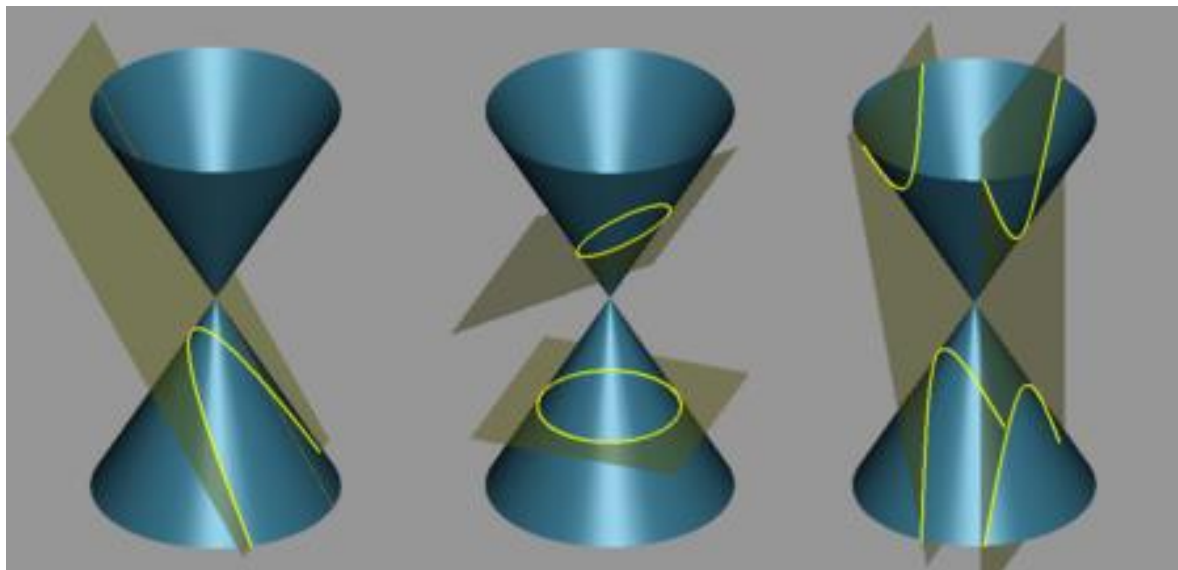
Local property of B-spline of degree 3

Agenda (omitted)

- 参数曲面表示
 - 参数表示的数学原理
 - 参数曲线
 - Bézier曲线
 - B-样条曲线
 - NURBS曲线
 - 参数曲面

引入NURBS曲线的原因

- B-样条情形不能精确表示二次曲面与平面的交线，如圆锥曲线(平面与圆锥的交线)



抛物线

椭圆(上)与圆(下)

双曲线

NURBS曲线

- NURBS (Non-Uniform Rational B-Spline):
非均匀有理B-样条的简称

- 定义：
$$\mathbf{R}(u) = \frac{\sum_{i=0}^n \omega_i \mathbf{R}_i N_{i,k}(u)}{\sum_{i=0}^n \omega_i N_{i,k}(u)}$$

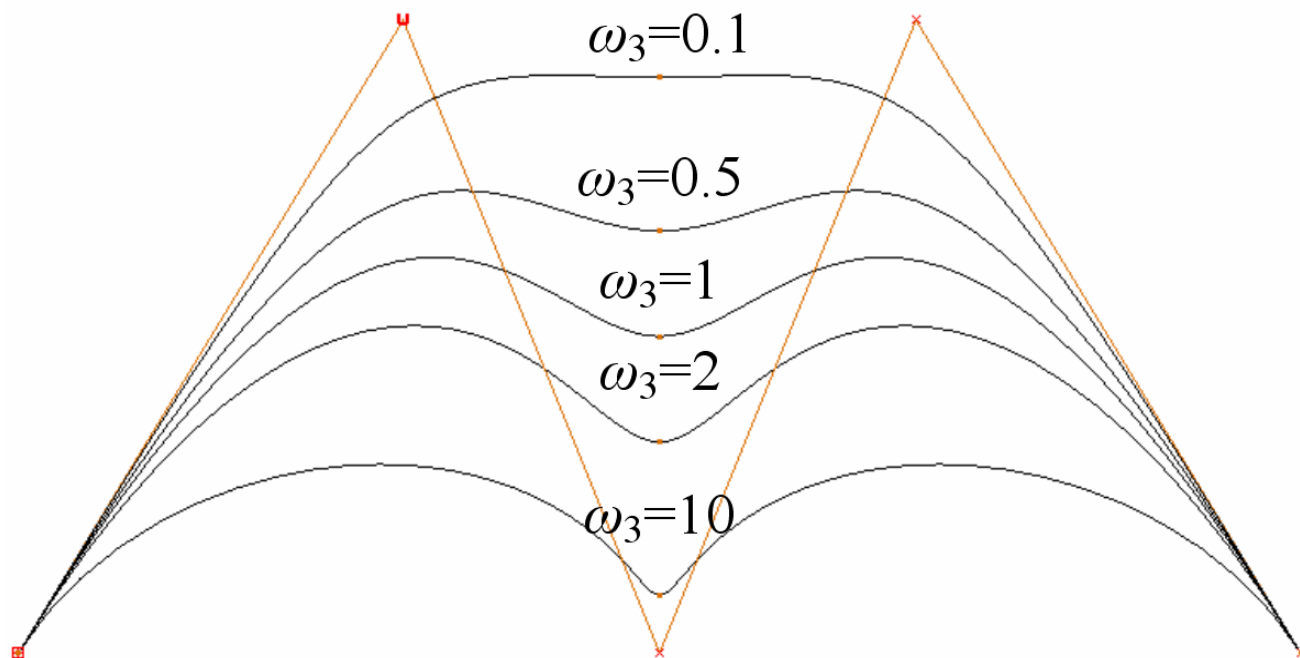
NURBS曲线

- $\{N_{i,k}(u)\}$ 为单位化的B-样条基函数
- $\{\mathbf{R}_i\}$ 为控制顶点
- NURBS曲线新增加的曲线控制手段是权因子 $\{\omega_i\}$,
 - 首末两个权因子 $\omega_0 > 0$ 、 $\omega_n > 0$
 - 其余的权因子满足 $\omega_i \geq 0$

NURBS曲线的权因子

- 每一个权因子对应于一个控制顶点
- 调整权因子的大小可以调整曲线的形状。
 - 当所有的权因子 $\omega_i=1$ 时，就是B-样条曲线；
 - 当某个权因子 $\omega_i=0$ 时，对应的控制顶点对曲线的形状没有影响
 - 当 $\omega_i \rightarrow \infty$ 时，曲线 $\mathbf{R}(u) \rightarrow \mathbf{R}_i$ ，即曲线过点 $\mathbf{R}_i$

NURBS曲线的例子



NURBS曲线权因子对曲线形状的影响

NURBS曲线表示圆

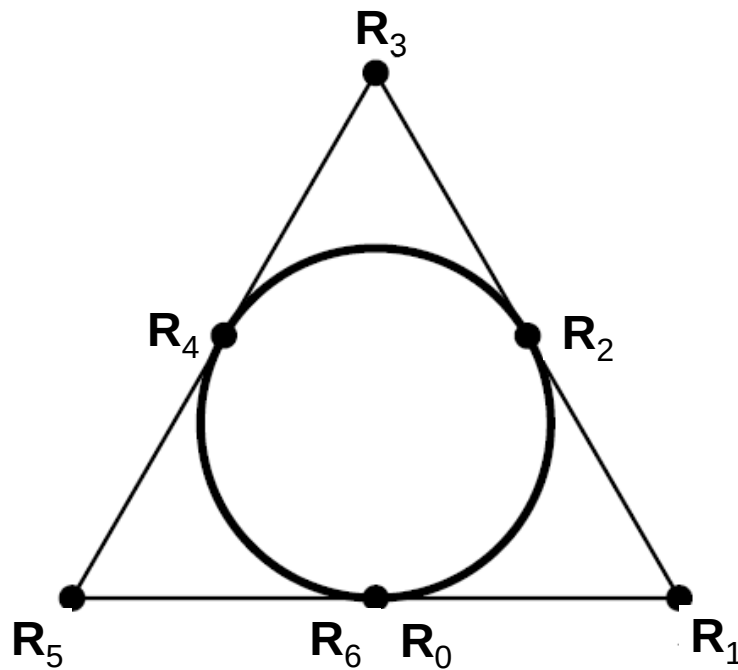
用三个 120° 圆弧表示圆：

$$\mathbf{u} = [0 \ 0 \ 0 \ 1 \ 1 \ 2 \ 2 \ 3 \ 3 \ 3]$$

$$k = 3$$

$$[\omega_i] = [1, \frac{1}{2}, 1, \frac{1}{2}, 1, \frac{1}{2}, 1]$$

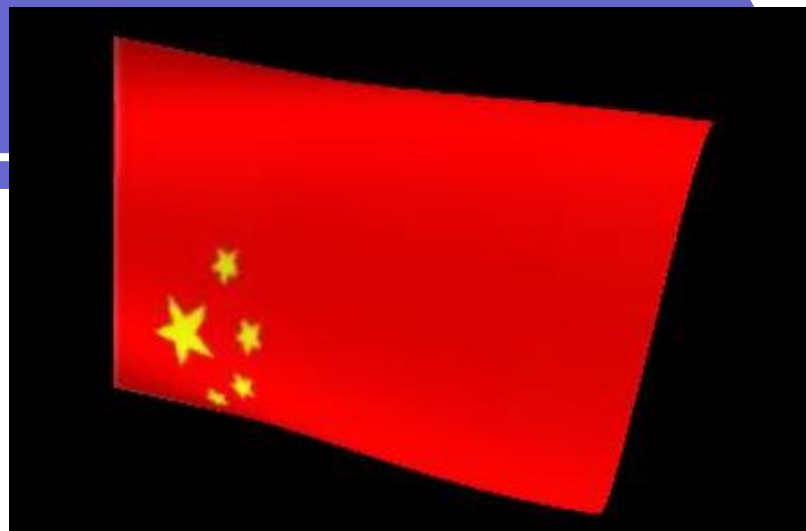
控制顶点分布如右图所示



NURBS曲线表示圆

Agenda

- Parametric curves and surfaces
 - 参数表示的数学原理
 - 参数曲线
 - Parametric surfaces
 - Bézier surfaces
 - B-样条曲面
 - NURBS曲面



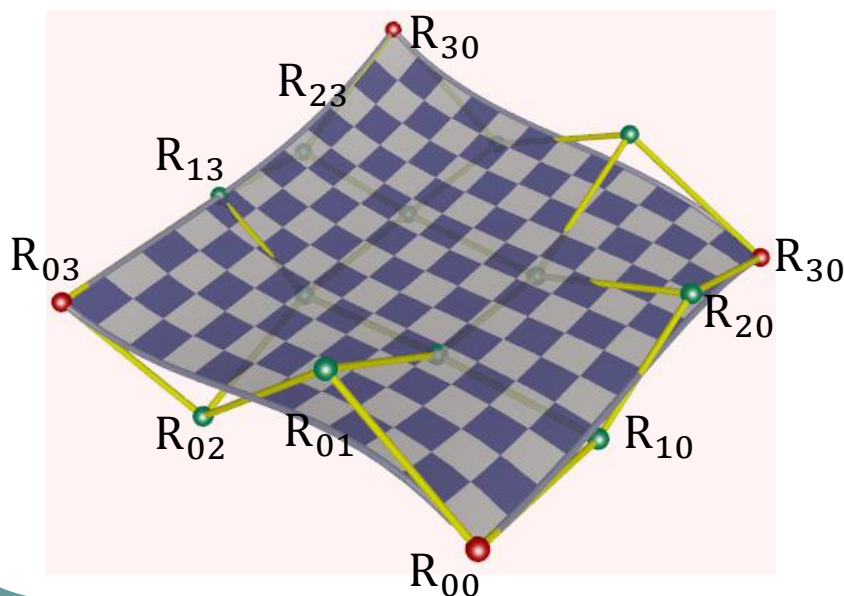
双三次Bézier曲面(Bicubic Bézier surfaces)

- 控制网格

$$R_{ij} = (x_{ij}, y_{ij}, z_{ij}), 0 \leq i, j \leq 3$$

- 曲面表示

$$R(u, v) = \sum_{i=0}^3 \sum_{j=0}^3 R_{ij} B_{i3}(u) B_{j3}(v), u, v \in [0, 1]$$



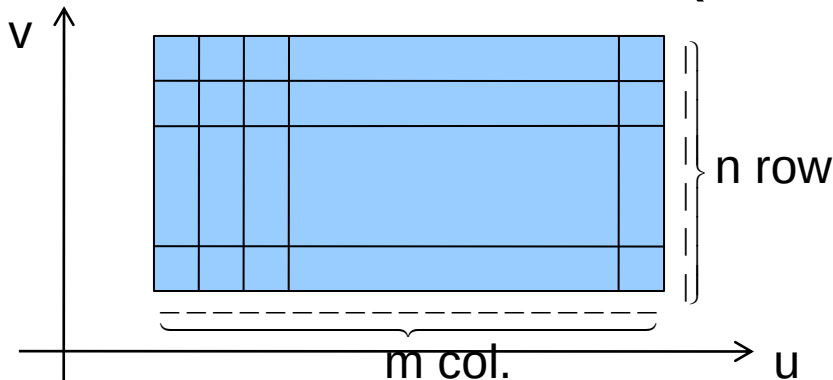
双三次Bézier曲面实例

Bézier曲面

- $m \times n$ 次Bézier曲面：

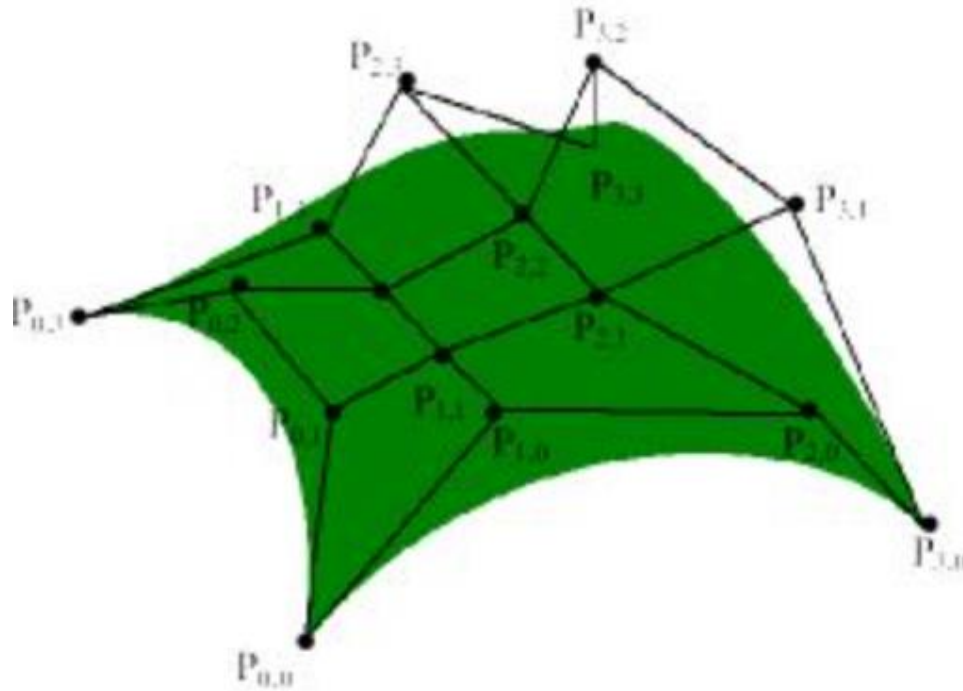
$$\mathbf{R}(u, v) = \sum_{i=0}^m \sum_{j=0}^n \mathbf{R}_{ij} B_{i,m}(u) B_{j,n}(v)$$

- $B_{i,m}(u)$ & $B_{j,n}(v)$: Bernstein基(bases)
- $\{\mathbf{R}_{ij}\}$ 规则连接形成控制网(Control mesh)



Bézier surfaces'性质

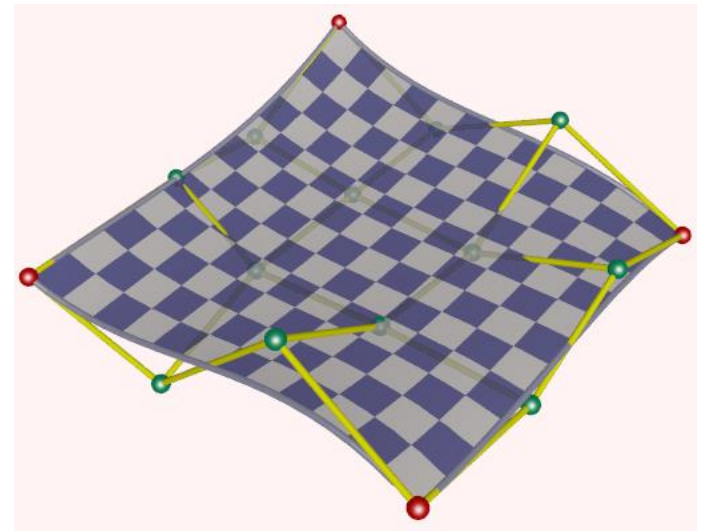
- The control mesh constrains the coarse shape of Bézier surfaces



Continued

- 插值角点(Interpolate four corner vertices)

$$\begin{array}{ll} \mathbf{R}(0,0) = \mathbf{R}_{00} & \mathbf{R}(1,0) = \mathbf{R}_{m0} \\ \mathbf{R}(0,1) = \mathbf{R}_{0n} & \mathbf{R}(1,1) = \mathbf{R}_{mn} \end{array}$$



Continued

- 角点处切向量计算(Tangents at four corner vertices)

$$\mathbf{R}_u(0,0) = m(\mathbf{R}_{10} - \mathbf{R}_{00})$$

$$\mathbf{R}_v(0,0) = n(\mathbf{R}_{01} - \mathbf{R}_{00})$$

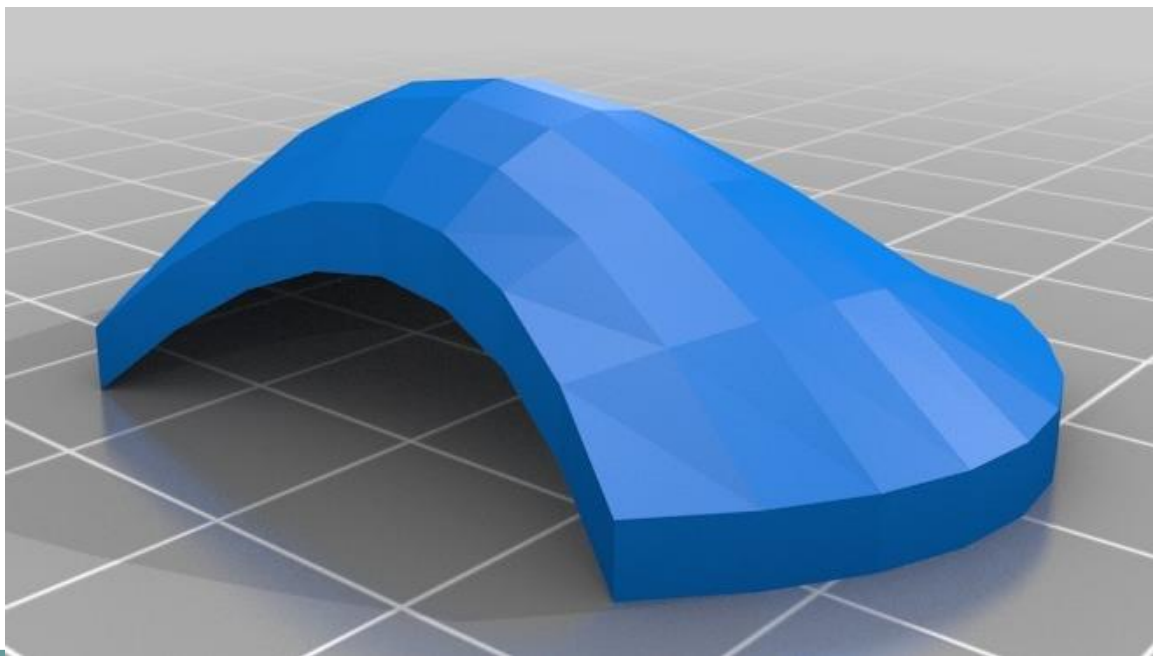
- Bézier曲面细分(Subdivision of Bézier surfaces)
 - 用加密的控制网格来逼近Bézier曲面

曲面绘制 (Rendering of Bézier surfaces)

```
Vector3D P[][]={{},...,{}}; // Control mesh
MyBezSurfDisplay()
{ ...;float step = 0.01;
    glBegin(GL_QUADS);
    for (u = 0; u <1; u+=step)
        for (v = 0; v<1; v+= step){
            ver0 = Bezier(u,v); ver1 = Bezier(u+step, v);
            ver2 = Bezier(u+step,v+step); ver1 = Bezier(u,v+step);
            计算上述4个点的法向n0,n1,n2,n3;
            glNormal3fv(n0); glVertex3fv(v0); glNormal3fv(n1); glVertex3fv(v1);
            glNormal3fv(n2); glVertex3fv(v2); glNormal3fv(n3); glVertex3fv(v3);
        }
    glEnd(); ...;
}
```

Bézier surfaces的缺点

- 全局性(Global property)
 - 改变一个控制顶点位置影响整个曲面形状
- 光滑拼接困难(smoothly merge multiple Bezier patches)



内容

- 参数曲面表示
 - 参数表示的数学原理
 - 参数曲线
 - 参数曲面
 - Bézier曲面
 - B-样条曲面
 - NURBS曲面

B样条曲面(B-spline surfaces)

- 定义

- 次数: $k_u + k_v$
- 控制顶点数: $(n_u + 1) \times (n_v + 1)$
- 节点向量(Knot vectors)

$$\mathbf{u} = \{u_0, u_1, \dots, u_i, \dots, u_{n_u + k_u + 1}\}$$

$$\mathbf{v} = \{v_0, v_1, \dots, v_j, \dots, v_{n_v + k_v + 1}\}$$

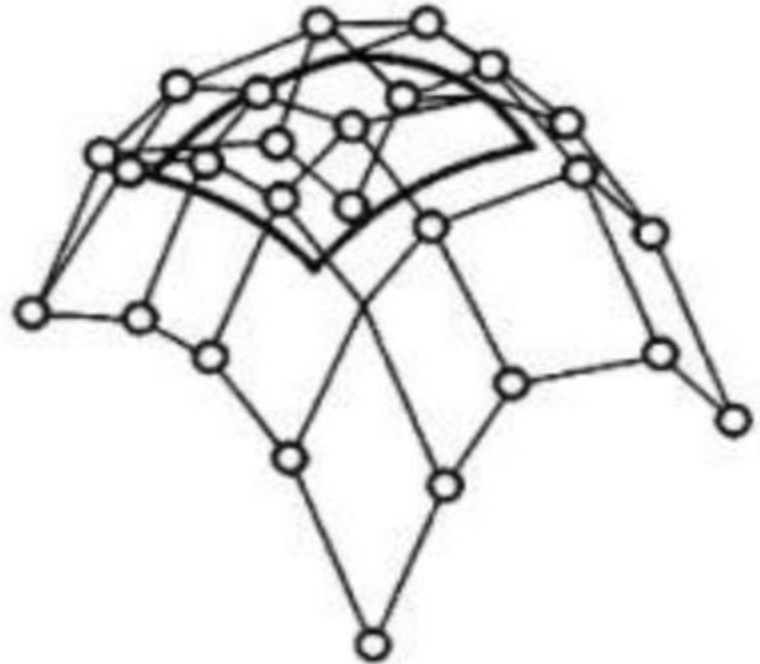
B-spline surfaces (B样条曲面)

- The surface (定义成张量积)

$$\mathbf{R}(u, v) = \sum_{i=0}^{n_u} \sum_{j=0}^{n_v} \mathbf{R}_{ij} N_{i,k_u}(u) N_{j,k_v}(v)$$

$\{\mathbf{R}_{ij}\}$: 控制网格(control net)

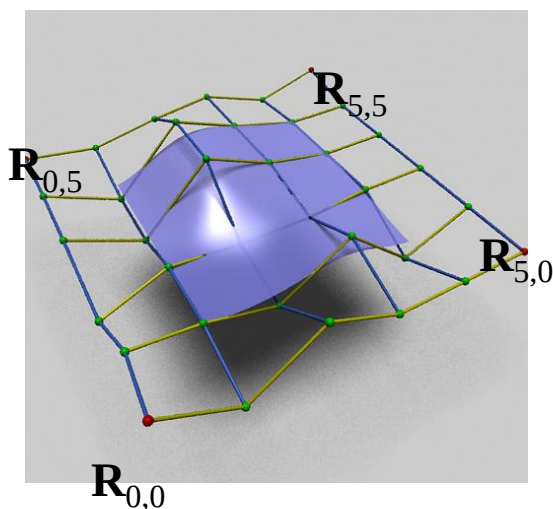
$N_{i,k_u}(u)$ 和 $N_{j,k_v}(v)$: B-spline 基



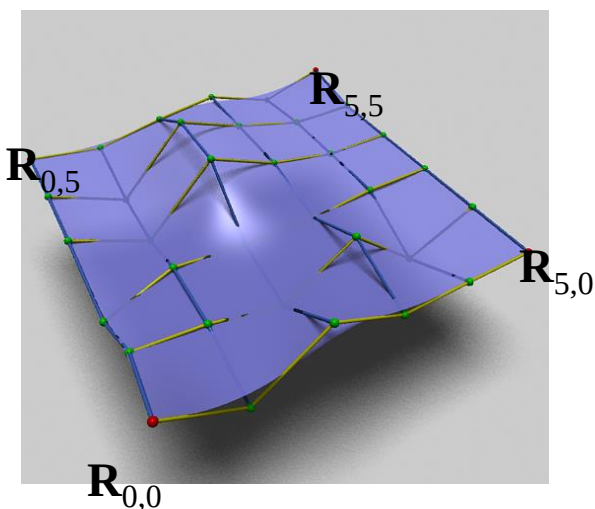
性质

- 局部性
- 控制顶点数
 - 次数确定，控制顶点数可任意
(Bézier曲面的次数确定后，控制顶点数目就定了)
- 其性质与曲线类似

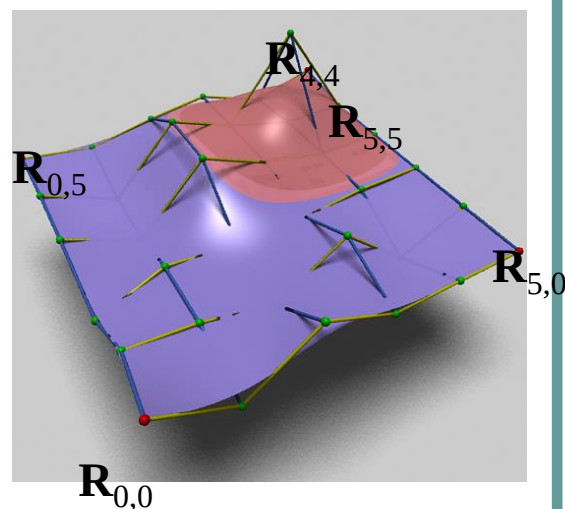
B-spline surfaces: examples



(a) Uniform knots



(b) 端点重节点

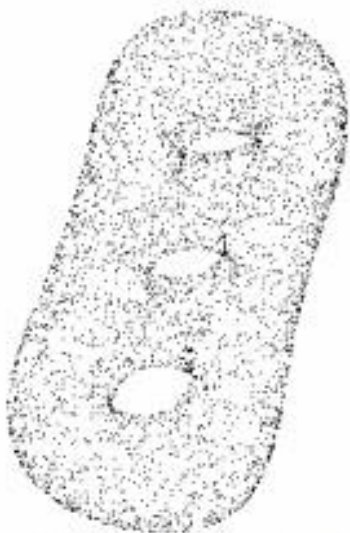


(c) Local property of B-spline surface

具有 6×6 个控制顶点双三次B-样条曲面：

- (a) 均匀节点向量 $\mathbf{u} = \mathbf{v} = [-4, -3, -2, -1, 0, 1, 2, 3, 4, 5]$ ，所构造曲面不插值角点
- (b) 具有端点处4阶重节点的节点向量 $\mathbf{u} = \mathbf{v} = [0, 0, 0, 0, 1, 2, 3, 3, 3, 3]$ ，曲面插值角点
- (c) 采用了与图(b)相同的节点向量，扰动顶点 $\mathbf{R}_{4,4}$ 的位置后，其形状变化的红色区域局限于变动顶点的邻域中。

Reconstruction(重构, Eck & Hoppe)



(a) Input: 4000 unorganized points P



(b) Step 1: Reconstructed mesh M_0



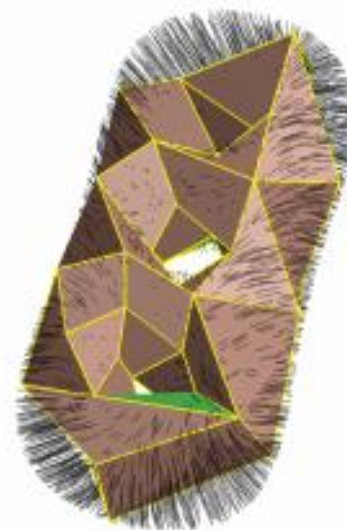
(c) Step 2a: Voronoi partition of M_0



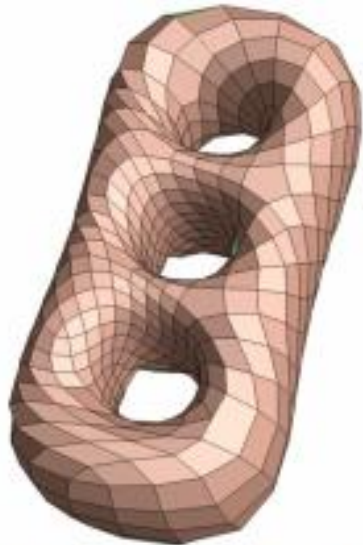
(d) Step 2b: Delaunay triangulation of M_0



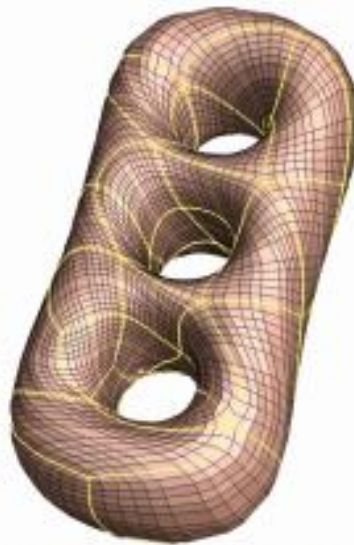
(e) Step 2c: Triangular base complex K_Δ



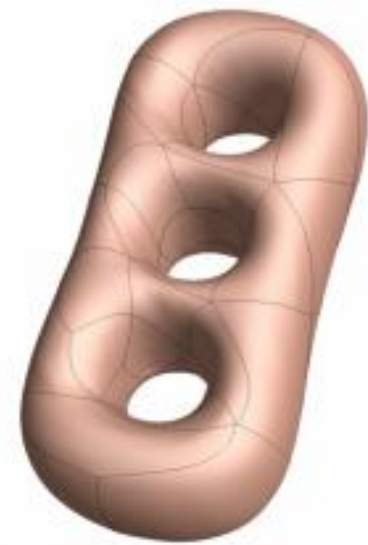
(f) Step 3: Quadrilateral base complex $K_\square$



(g) Step 4: Optimized control mesh M_k



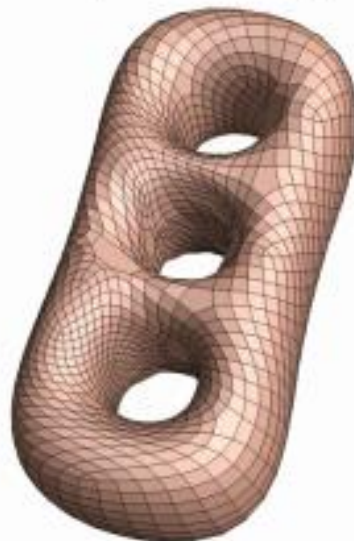
(h) Step 4: B-spline control net $d_{k,s}$



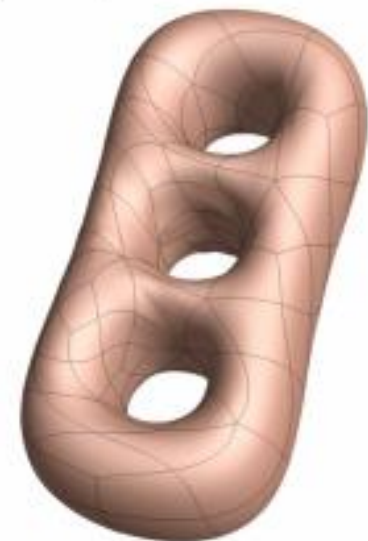
(i) Step 4: B-spline surface S



(j) Step 5: Adaptively refined K_Θ



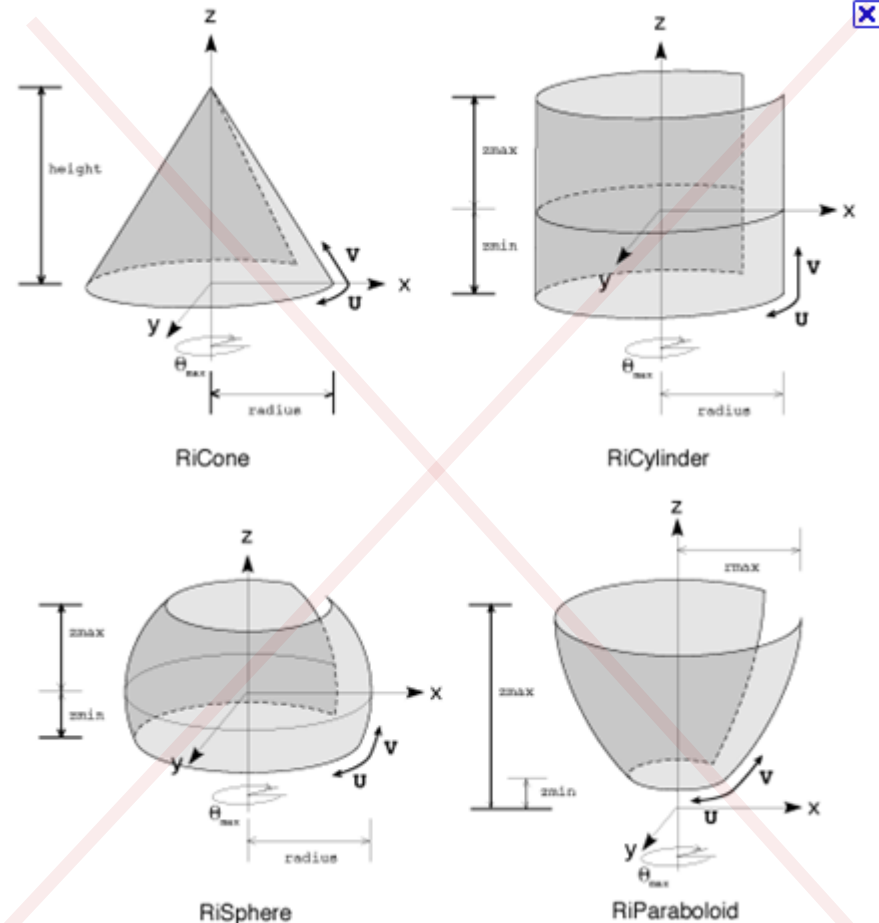
(k) Step 5: Optimized control mesh M_k



(l) Step 5: Final B-spline surface S

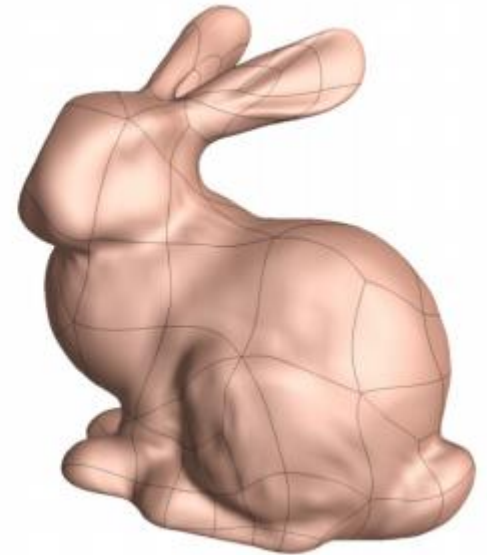
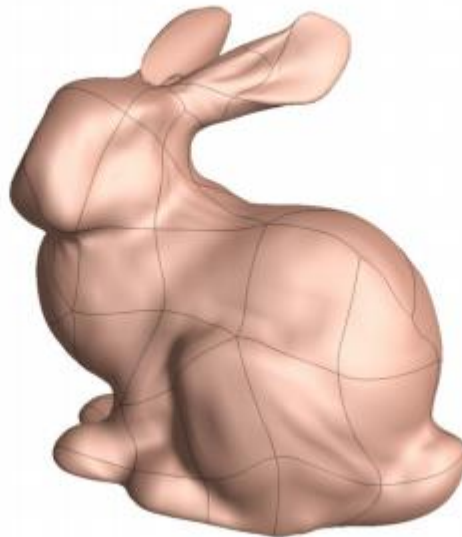
Drawbacks of B-spline surfaces

- It can not exactly represent conic surfaces
(不能精确表示二次曲面：球面、圆柱面圆锥面等)

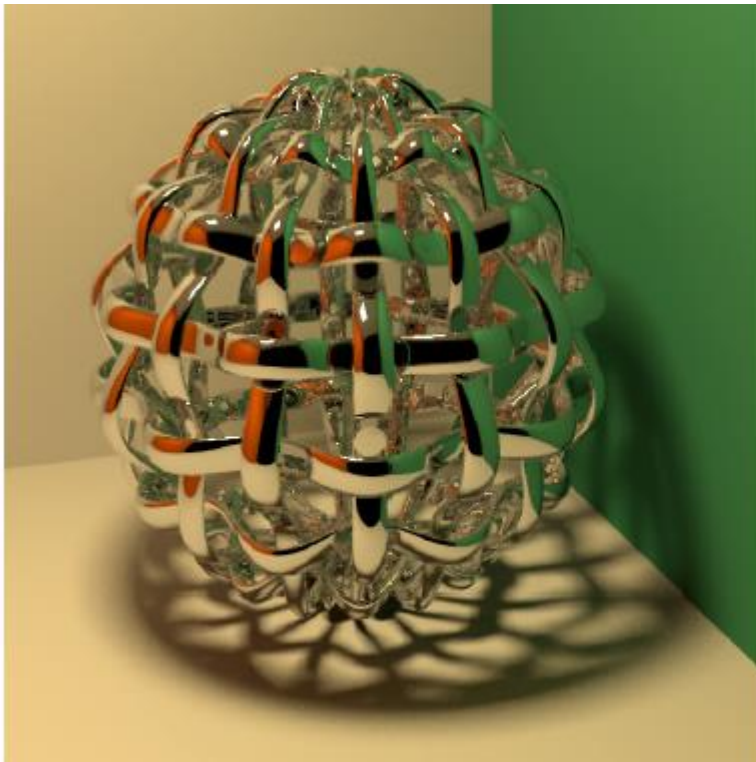


Drawbacks of B-spline surfaces

- Smooth stitching is also difficult(光滑拼接仍然困难)



- It is more difficult for complicated objects



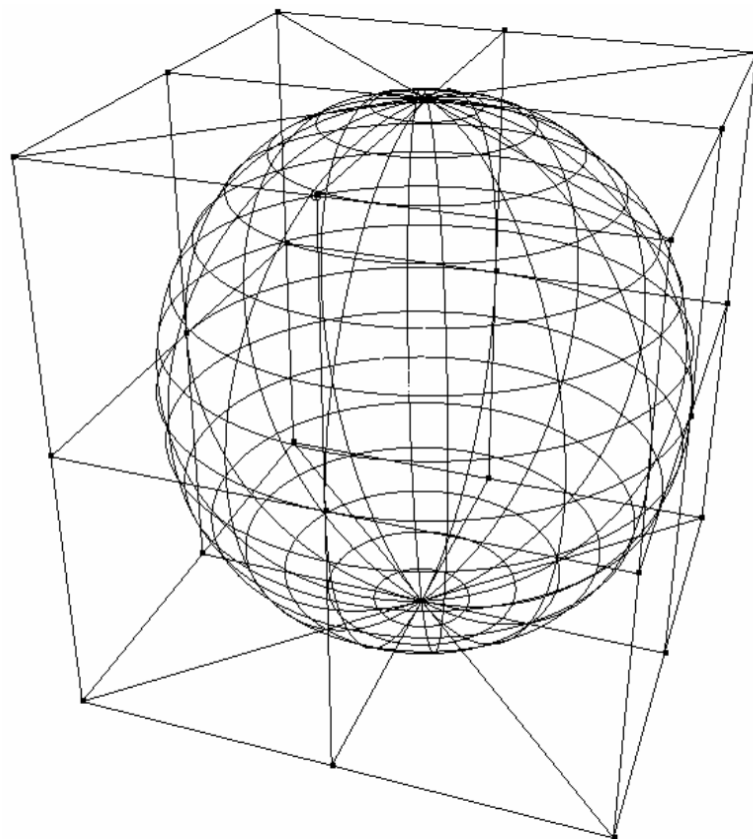
内容

- 参数曲面表示
 - 参数表示的数学原理
 - 参数曲线
 - 参数曲面
 - Bézier曲面
 - B-样条曲面
 - NURBS曲面

NURBS曲面(omitted)

- NURBS曲面
 - 增加了权因子作为形状控制手段
 - 包含B-样条曲面和Bézier曲面
 - 可以精确表示机械零件中常用的二次曲面
- 工业产品几何定义的STEP标准 (1991年):
 - 自由曲线曲面唯一地采用NURBS表示

NURBS曲面表示球面



NURBS精确表示的球面及其控制顶点

小结

- 物体的参数曲面表示
 - 参数表示的数学原理：曲线、曲面
 - 参数曲线：Bézier、B-样条和NURBS曲线
 - 参数曲面：Bézier、B-样条和NURBS曲面
 - 参数曲线与曲面的绘制